

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	8
MDMT Summary:	9
Warnings and Errors:	10
Input Echo:	11
XY Coordinate Calculations:	20
Internal Pressure Calculations:	21
Element: Bottom Clad Dis	22
Element: Clad shell 1	24
Element: Unclad Shell	26
Element: Unclad Topdish	28
Shop Pressure Test Results:	30
Field Pressure Test Results:	32
External Pressure Calculations:	34
Element and Detail Weights:	35
Nozzle Flange MAWP:	38
Natural Frequency Calculation:	40
Wind Load Calculation:	41
Earthquake Load Calculation:	44
Wind/Earthquake Shear, Bending:	46
Wind/Earthquake Deflection:	47
Longitudinal Stress Constants:	48
Longitudinal Allowable Stresses:	49
Longitudinal Stresses due to:	50
Stress due to Combined Loads:	52
Skirt Stress Calculations:	57
Center of Gravity Calculation:	62
Lifting Lug Calcs: lifting lug	64
Basering Calculations:	72
Nozzle Calcs.: D1-2"	77
Nozzle Calcs.: K1-2"	82
Nozzle Calcs.: K3-2"	85
Nozzle Calcs.: K5-2"	88
Nozzle Calcs.: K7-2"	91
Nozzle Calcs.: K2-2"	94
Nozzle Calcs.: K4-2"	97
Nozzle Calcs.: K6-2"	100
Nozzle Calcs.: K8-2"	103
Nozzle Calcs.: K10-2"	106
Nozzle Calcs.: N1-24"	109
Nozzle Calcs.: N2-24"	123
Nozzle Calcs.: M1-24"	137
Nozzle Calcs.: K9-2"	143



Process Air 3rd KOD 198-D

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Process Air Compressor 3rd KO Drum-198-D-7%.pvdb

Date of Analysis : Sep 16, 2026 2:30pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2906.000 mm. ID
 Vessel Design Length, Tangent to Tangent 3125.00 mm.
 Distance of Bottom Tangent above Grade 3000.00 mm.
 Distance of Base above Grade 300.00 mm.
 Specified Datum Line Distance 2700.00 mm.
 Internal Design Temperature 82 °C
 Internal Design Pressure 25.000 bars
 External Design Temperature 0 °C
 Maximum Allowable Working Pressure 6 bars
 Shop Test Pressure 34.120 bars
 Field Test Pressure 32.500 bars
 Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -17.0 °C
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-36	...	...	K02600	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-312 TP316L	...	...	S31603	
Nozzle	SA-182 F316L	...	<= 130	S31603	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-182 F316L	...	<= 130	S31603	
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt2	...	...	...	1.0000	N/A	No
Skirt1	...	...	...	1.0000	N/A	No
Bottom Clad Dis	25.141	0.00	25.500	1.0000	No	No
Clad shell 1	25.065	0.00	27.100	1.0000	N/A	No
Unclad Shell	25.007	0.00	27.200	1.0000	N/A	No
Unclad Topdish	25.007	0.00	25.600	1.0000	No	No

Liquid Level: 1376.50 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.991

Element Types and Properties:

Element	"To" Elev	Element	Nominal	Finished	Reqd Thk	Reqd Thk	Long	Circ
---------	-----------	---------	---------	----------	----------	----------	------	------

Type	mm.	Length mm.	Thickness mm.	Thickness mm.	Internal mm.	External mm.	Eff	Eff
Skirt	-1000.0	1700.0	12.0	12.0	...	...	0.70	0.70
Skirt	0.0	1000.0	10.0	10.0	...	...	0.70	0.50
Ellipse	50.0	50.0	34.0	28.0	27.5	...	1.00	1.00
Cylinder	950.0	900.0	30.0	30.0	27.7	...	1.00	1.00
Cylinder	3075.0	2125.0	30.0	30.0	27.6	...	1.00	1.00
Ellipse	3125.0	50.0	34.0	28.0	27.4	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Lift Lug	WRC-107/537	0.445	n/a	Passed
D1-2"	WRC-107/537	0.216	n/a	Passed
N1-24"	WRC-297	0.473	Nozzle	Passed
N2-24"	WRC-297	0.489	Nozzle	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	850	1808.83	6592.22	7448.8	29508.6
20	30	2200	1558.6	6488.72	4586.43	18389.5
30	40	2725	1411.51	6412.55	3101.34	11938.8
40	50	3200	1148.45	5070.1	3037.34	11651.7
50	60	4712.5	1015.21	4155.62	2063.68	7500.06
60	70	5800	700.621	2194.08	240.568	753.368

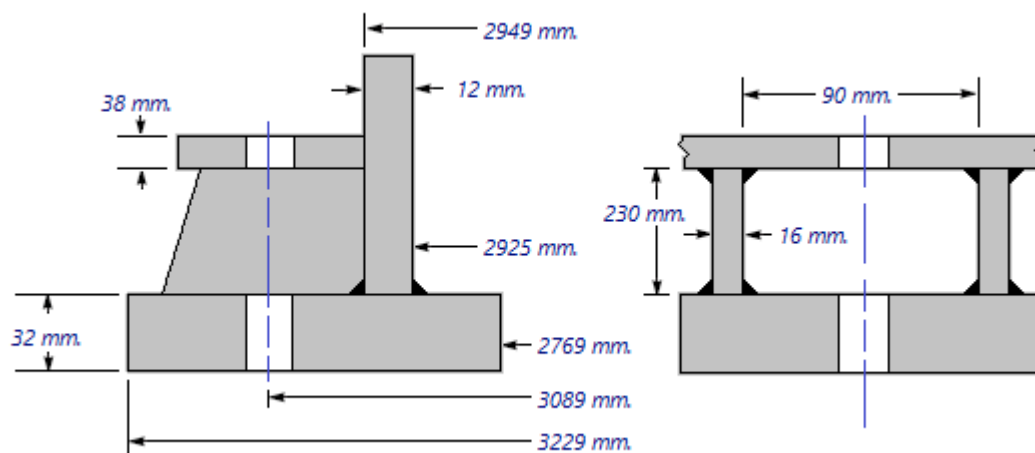
Abs Max of the all of the Stress Ratio's : 0.4820

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing 32.0000 mm.
 Inside Diameter of Basing 2769.0000 mm.
 Outside Diameter of Basing 3228.9998 mm.
 Nominal Diameter of Bolts 42.0000 mm.
 Diameter of Bolt Circle 3088.9998 mm.
 Number of Bolts 16

Thickness of Gusset Plates 16.0000 mm.
 Average Width of Gusset Plates 130.0000 mm.
 Height of Gussets 230.0000 mm.
 Distance between Gussets 90.0000 mm.
 Thickness of Top Plate or Ring 38.0000 mm.
 Radial Width of the Top Plate 140.0000 mm.
 Number of Gussets 32

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	1809. Kg
Total Earthquake Shear on Support	6592. Kg
Wind Moment on Support	7599. Kg-m.
Earthquake Moment on Support	29659. Kg-m.
Maximum Initial Bolt Load	86. Kg

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	21863.3 kg.
Shop Test - Fabricated + water (full)	49153.1 kg.
Shipping - Fab. + rem. intls.+ shipping app.	21863.3 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	30804.9 kg.
Empty - Fab. + intls. + details + wghts.	30804.9 kg.
Operating - Empty + operating liquid (no ca)	38220.2 kg.
Field Test - Empty weight + test medium (full)	57798.7 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
D1-2"	6	...	...	OK	3.42	OK	No Calc[*]
K1-2"		...	...	OK	4.80	OK	No Calc[*]
K3-2"		...	...	OK	4.80	OK	No Calc[*]
K5-2"		...	...	OK	4.80	OK	No Calc[*]
K7-2"		...	...	OK	4.80	OK	No Calc[*]
K2-2"		...	...	OK	4.80	OK	No Calc[*]
K4-2"		...	...	OK	4.80	OK	No Calc[*]
K6-2"		...	...	OK	4.80	OK	No Calc[*]
K8-2"		...	...	OK	4.80	OK	No Calc[*]
K10-2"		...	...	OK	4.80	OK	No Calc[*]
N1-24"		...	...	OK	7.35	OK	No Calc[*]
N2-24"		...	...	OK	7.35	OK	No Calc[*]
M1-24"		...	...	...		OK	No Calc[*]
K9-2"		...	...	OK	5.80	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 6 Nozzle : D1-2"
Minimum MAWP Shells/Flanges : 6 Element : Bottom Clad Dis
Minimum MAPnc Shells/Flanges : 26.5 Element : Unclad Topdish

Computed Vessel M.A.W.P. : 6 bars

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	D1-2"	0.000	0.000	114.325
40	K1-2"	850.000	90.000	142.328
40	K3-2"	850.000	80.000	142.328
40	K5-2"	850.000	100.000	142.328
40	K7-2"	850.000	270.000	142.328
40	K2-2"	150.000	80.000	142.328
40	K4-2"	150.000	90.000	142.328
40	K6-2"	150.000	100.000	142.328
40	K8-2"	150.000	110.000	142.328
40	K10-2"	150.000	270.000	142.328
50	N1-24"	1750.000	90.000	1175.200
50	N2-24"	2350.000	270.000	1167.200
50	M1-24"	1500.000	180.000	1167.200
50	K9-2"	2500.000	110.000	142.328

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
D1-2"	2.000 in	80S	WNF	2.375	5.537	...	...	178.17	600
K1-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	236.99	300
K3-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	236.99	300
K5-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	236.99	300
K7-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	236.99	300
K2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	236.99	300
K4-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	236.99	300
K6-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	236.99	300
K8-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	236.99	300
K10-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	236.99	300
K9-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	236.99	300
N1-24"	24.000 in	Actual	WNF	24.000	14.000	1109.60	32.00	362.33	300
N2-24"	24.000 in	Actual	WNF	24.000	16.000	1109.60	32.00	362.33	300
M1-24"	24.000 in	Actual	WNF	24.000	16.000	1109.60	32.00	362.33	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
D1-2"	SA-312 TP316L	28.000	10.000	...	...	...
K1-2"	SA-182 F316L	30.000	10.000	...	...	...
K3-2"	SA-182 F316L	30.000	10.000	...	...	...
K5-2"	SA-182 F316L	30.000	10.000	...	...	...
K7-2"	SA-182 F316L	30.000	10.000	...	...	...
K2-2"	SA-182 F316L	30.000	10.000	...	...	...
K4-2"	SA-182 F316L	30.000	10.000	...	...	...
K6-2"	SA-182 F316L	30.000	10.000	...	...	...
K8-2"	SA-182 F316L	30.000	10.000	...	...	...
K10-2"	SA-182 F316L	30.000	10.000	...	...	...
K9-2"	SA-105	30.000	10.000	...	...	...
N1-24"	SA-516 70	32.000	10.000	24.000	32.000	...
N2-24"	SA-516 70	32.000	10.000	24.000	32.000	...
M1-24"	SA-516 70	32.000	10.000	24.000	32.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
D1-2"	...	0.0	150.00	0.00	Bottom Clad Dis
K1-2"	850.000	90.0	206.38	0.00	Clad shell 1
K3-2"	850.000	80.0	206.38	0.00	Clad shell 1
K5-2"	850.000	100.0	206.38	0.00	Clad shell 1
K7-2"	850.000	270.0	206.38	0.00	Clad shell 1
K2-2"	150.000	80.0	206.38	0.00	Clad shell 1
K4-2"	150.000	90.0	206.38	0.00	Clad shell 1
K6-2"	150.000	100.0	206.38	0.00	Clad shell 1

Process Air 3rd KOD 198-D

PV Elite 28 U1 Licensee: SL User



FileName : Process Air Compressor 3rd KO Drum-198-D-7%

Page 7 of 145

Nozzle Schedule:

Step: 41 2:30pm Sep 16,2026



HEXAGON

K8-2"	150.000	110.0	206.38	0.00	Clad shell 1
K10-2"	150.000	270.0	206.38	0.00	Clad shell 1
K9-2"	2500.000	110.0	206.38	0.00	Unclad Shell
N1-24"	1750.000	90.0	300.00	0.00	Unclad Shell
N2-24"	2350.000	270.0	300.00	0.00	Unclad Shell
M1-24"	1500.000	180.0	300.00	0.00	Unclad Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 34.0mm. THK X 2906.0mm. ID X 50.0mm.	SA-516 70
1	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 3000.0mm	...
2	CLADDING: 776.500mm X 3.000mm THK	...
1	CLADDING: 900.000mm X 3.000mm THK	...
1	CLADDING: 2125.000mm X 3.000mm THK	...
2	LIFTING LUGS:	SA-516 70
2	LIFTING LUG PADS: 480.0mm X 480.0mm X 32.0mm THK	
1	PLATFORM: 1000mm X 5000mm WIDE	
1	LADDER: 3800.0mm	...
1	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 3229.0mm. OD X 2769.0 ID X 32.0mm.	SA-516 70
16	BASERING BOLTS: 42.0mm. DIA	SA-193 B7
16	BASERING BOLT NUTS: 42.0mm. DIA	...
1	BASERING TOP RING: 3229.0mm. OD X 2949.0 ID X 38.0mm.	...
32	BASERING GUSSET PLATES: 130.0mm. X 130.0mm. X 230.0mm. X 16.0mm.	...
2	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 3019.6mm. X 1700.0mm.	SA-36
1	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 1509.8mm. X 1700.0mm.	SA-36
1	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 1640.1mm. X 1700.0mm.	SA-36
2	CYLINDRICAL SKIRT SEGMENT 10.0mm. THK X 3021.6mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 10.0mm. THK X 1510.8mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 10.0mm. THK X 1641.2mm. X 1000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 30.0mm. THK X 3000.0mm. X 900.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 30.0mm. THK X 1500.0mm. X 900.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 30.0mm. THK X 1629.4mm. X 900.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 30.0mm. THK X 3000.0mm. X 2125.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 30.0mm. THK X 1500.0mm. X 2125.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 30.0mm. THK X 1629.4mm. X 2125.0mm.	SA-516 70
1	NAMEPLATE	...


Minimum Design Metal Temperature Results Summary (°C):

Description Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Bottom Clad D[10]	D	-32	-104		0.234	28.000	1.00	No
Bottom Clad Di[7]	D	-28	-104		0.192	34.000	1.00	No
Clad shell 1 [20]	D	-30	-48		0.462	30.000	1.00	No
Unclad Shell [20]	D	-30	-48		0.462	30.000	1.00	No
Unclad Topdis[10]	D	-32	-104		0.234	28.000	1.00	No
Unclad Topdish[7]	D	-28	-104		0.192	34.000	1.00	No
N1-24" [1]	D	2	-104		0.120	14.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
N2-24" [1]	D	6	-104		0.102	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
M1-24" [1]	D	6	-104		0.102	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K9-2" [1]	D	-44	-104	-29	0.007	16.764	1.00	No
Nozzle Flg [5]	!	-17	-17					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Warmest MDMT:		6	-17					

Required Minimum Design Metal Temperature -12.0 °C

Warmest Computed Minimum Design Metal Temperature -17.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 20 40 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Process Air 3rd KOD 198-D

Design Internal Pressure (for Hydrotest) 25 bars
 Design Internal Temperature 82.0 °C
 Projection of Nozzle from Vessel Top 500 mm.
 Projection of Nozzle from Vessel Bottom 500 mm.
 Minimum Design Metal Temperature -12.0 °C
 Type of Construction Welded
 Special Service None
 Degree of Radiography RT-1
 Use Higher Longitudinal Stresses (Flag) Y
 Select t for Internal Pressure (Flag) N
 Select t for External Pressure (Flag) N
 Select t for Axial Stress (Flag) N
 Select Location for Stiff. Rings (Flag) N
 Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test UG-99(c)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1 NP+EW+1.1WI+FW+BW
 Load Case 2 NP+EW+EE+FS+BS
 Load Case 3 NP+OW+1.1WI+FW+BW
 Load Case 4 NP+OW+EQ+FS+BS
 Load Case 5 NP+HW+HI
 Load Case 6 NP+HW+HE
 Load Case 7 IP+OW+1.1WI+FW+BW
 Load Case 8 IP+OW+EQ+FS+BS
 Load Case 9 EP+OW+1.1WI+FW+BW
 Load Case 10 EP+OW+EQ+FS+BS
 Load Case 11 HP+HW+HI
 Load Case 12 HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.050
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

**Complete Listing of Vessel Elements and Details:**

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt2
Distance "FROM" to "TO"	1700 mm.
Skirt Outside Diameter	2949 mm.
Diameter of Skirt at Base	2949 mm.
Skirt Thickness	12 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.3099999
Material Name	SA-36
Allowable Stress, Ambient	114 N./mm2
Allowable Stress, Operating	114 N./mm2
Allowable Stress, Hydrotest	225 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	230.08 N./mm2
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	K02600
Product Form	Plate; sheet
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt1
Distance "FROM" to "TO"	1000 mm.
Skirt Outside Diameter	2947 mm.
Diameter of Skirt at Base	2947 mm.
Skirt Thickness	10 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	10 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.3099999
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	138 N./mm2
Allowable Stress, Hydrotest	234 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	242.6 N./mm2
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No

Element#	3/6
Element From Node	30
Element To Node	40



Element Type	Elliptical
Description	Bottom Clad Dis
Distance "FROM" to "TO"	50 mm.
Inside Diameter	2906 mm.
Element Thickness	28 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	34 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	25 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.3099999
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	360.0
Platform End Angle (degrees)	0.0
Platform Wind Area	30000 cm2
Platform Weight	3823.6 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm2
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	85 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	90.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	3000 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	0 mm.

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	-726.5 mm.
Height/Length of Liquid	776.5 mm.
Liquid Density	0.0009894 kg./cm3

Element From Node	30
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	-726.5 mm.
Height/Length of Lining	776.5 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm3

Element From Node	30
Detail Type	Nozzle
Detail ID	D1-2"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	80S
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-312 TP316L



Element#	4/6
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Clad shell 1
Distance "FROM" to "TO"	900 mm.
Inside Diameter	2906 mm.
Element Thickness	30 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	30 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	25 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.3099999
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	600 mm.
Liquid Density	0.0009894 kg./cm3
Element From Node	40
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	900 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm3
Element From Node	40
Detail Type	Nozzle
Detail ID	K1-2"
Dist. from "FROM" Node / Offset dist	800 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L
Element From Node	40
Detail Type	Nozzle
Detail ID	K3-2"
Dist. from "FROM" Node / Offset dist	800 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	80.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L
Element From Node	40
Detail Type	Nozzle
Detail ID	K5-2"
Dist. from "FROM" Node / Offset dist	800 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300



Layout Angle	100.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K7-2"	
Dist. from "FROM" Node / Offset dist	800	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K2-2"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	80.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K4-2"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K6-2"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	100.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K8-2"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	110.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	



Nozzle Matl	SA-182 F316L
Element From Node	40
Detail Type	Nozzle
Detail ID	K10-2"
Dist. from "FROM" Node / Offset dist	100 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L

Element From Node	40
Detail Type	Weight
Detail ID	nozzles lining
Dist. from "FROM" Node / Offset dist	-0.45720E-04 mm.
Miscellaneous Weight	20 Kg
Offset from Element Centerline	0 mm.

Element#	5/6
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Unclad Shell
Distance "FROM" to "TO"	2125 mm.
Inside Diameter	2906 mm.
Element Thickness	30 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	30 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	25 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.3099999
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	2125 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm3

Element From Node	50
Detail Type	Nozzle
Detail ID	N1-24"
Dist. from "FROM" Node / Offset dist	800 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	50
Detail Type	Nozzle
Detail ID	N2-24"



Dist. from "FROM" Node / Offset dist	1400	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	M1-24"	
Dist. from "FROM" Node / Offset dist	550	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	K9-2"	
Dist. from "FROM" Node / Offset dist	1550	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	110.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	50	
Detail Type	Weight	
Detail ID	Donut Ring	
Dist. from "FROM" Node / Offset dist	-0.0001463	mm.
Miscellaneous Weight	130	Kgf
Offset from Element Centerline	0	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	Demister pad	
Dist. from "FROM" Node / Offset dist	-0.0001463	mm.
Miscellaneous Weight	150	Kgf
Offset from Element Centerline	0	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	N1 Distributor	
Dist. from "FROM" Node / Offset dist	1000	mm.
Miscellaneous Weight	850	Kgf
Offset from Element Centerline	1000	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	N2-Hopper	
Dist. from "FROM" Node / Offset dist	1400	mm.
Miscellaneous Weight	700	Kgf
Offset from Element Centerline	1000	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	Davit	
Dist. from "FROM" Node / Offset dist	550	mm.
Miscellaneous Weight	50	Kgf

Offset from Element Centerline 1500 mm.

Element#	6/6	
Element From Node	60	
Element To Node	70	
Element Type	Elliptical	
Description	Unclad Topdish	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	2906	mm.
Element Thickness	28	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	34	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	25	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.3099999	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	837	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	65443	cm2
Platform Weight	4392.8	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	5000	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	2500	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	135.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	3800	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	5000	mm.
Element From Node	60	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Lining	776.5	mm.
Thickness of Lining	3	mm.
Density	0.008	kg./cm3
Element From Node	60	
Detail Type	Weight	
Detail ID	Clip weight	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	360	Kgf
Offset from Element Centerline	0	mm.

XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
	Skirt2	...	1700	...	1700
	Skirt1	...	2700	...	1000
	Bottom Clad Dis	...	2750	...	50
	Clad shell 1	...	3650	...	900
	Unclad Shell	...	5775	...	2125
	Unclad Topdish	...	5825	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
	Skirt2	...	12	1	2949	...	1
	Skirt1	...	10	1	2947	...	2
	Bottom Clad Dis	25.141	34	1	2906	138	3
	Clad shell 1	25.065	30	1	2906	138	4
	Unclad Shell	25.007	30	1	2906	138	5
	Unclad Topdish	25.007	34	1	2906	138	6

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
	Skirt2	...	No Calc	No Calc	12	No Calc	1
	Skirt1	...	No Calc	No Calc	10	No Calc	2
	Bottom Clad Dis	25	25.5	26.5	28	27.5144	3
	Clad shell 1	25	27.1	28.1	30	27.7019	4
	Unclad Shell	25	27.2	28.1	30	27.6391	5
	Unclad Topdish	25	25.6	26.5	28	27.3731	6

Minimum 6 26.5

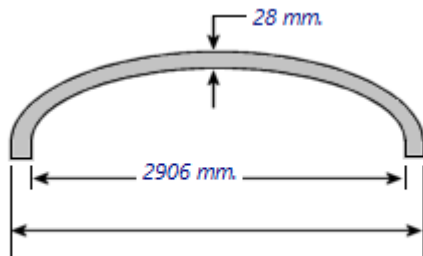
MAWP: 6 bars, limited by: DESIGN (user specified)

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom Clad Dis

Elliptical Head Design Information:					
Design Pressure	P	25.141 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2906.000 mm.	Actual Stress		135.5 N./mm^2
Spec. Min. Thk t _e , or s _a		28.0000 mm.	Spec. Nominal Thk.		34.000 mm.
Surface Area		100546.5 cm^2	SG of Contents		0.991
Element Volume		3544632.0 cm3	Weight of Contents		3489.5 Kg
Empty Weight		2847.7 Kg	Operating Weight		6337.2 Kg
			Comp. All. Ope. Stress		127.9 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 36.325 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.162 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (25.1 \cdot 2908 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 25.1)$$

$$= 26.5144 + 1.0000 = 27.5144 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.141 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 27) / (1 \cdot 2908 + 0.2 \cdot 27)$$

$$= 25.6 - 0.14 = 25.5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 28) / (1 \cdot 2906 + 0.2 \cdot 28)$$

$$= 26.5 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (25.1 \cdot (1 \cdot 2908 + 0.2 \cdot 27)) / (2 \cdot 1 \cdot 27)$$

$$= 135.523 \text{ N./mm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (25.1 \cdot 1454) / (138 \cdot 1 - 0.6 \cdot 25.1) + 1$
 $= 27.783 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.070 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 1 \cdot 33) / (1454 + 0.6 \cdot 33)$
 $= 30.9 - 0.07 = 30.8 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$
 $= (2 + (2908 / (2 \cdot 727)))^2 / 6$
 $= 0.999084$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.990 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 28$, $t_r = 6.32$, $c = 1 \text{ mm.}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.23$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -32 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 34$, $t_r = 6.34$, $c = 1 \text{ mm.}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.19$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -28 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	6.000 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	28.000 mm.	Basic MDMT	-32 °C
Stress/Thk. Ratio	0.234	Temperature Difference	72 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

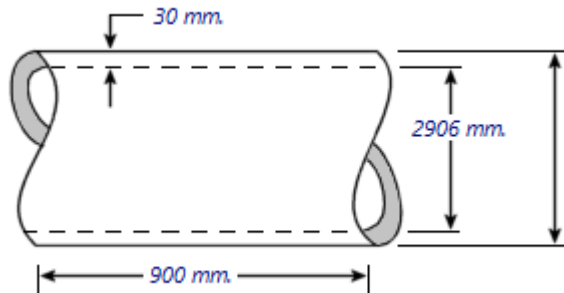
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 36.325 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.162 mm.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Clad shell 1

Cylindrical Shell Design Information:					
Design Pressure	P	25.065 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		900.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2906.000 mm.	Actual Stress		127.2 N./mm^2
Spec. Min. Thk t _e , or s _a		30.0000 mm.	Spec. Nominal Thk.		30.000 mm.
Surface Area		83861.7 cm^2	SG of Contents		0.991
Element Volume		5970372.5 cm3	Weight of Contents		3925.8 Kg
Empty Weight		1930.1 Kg	Operating Weight		5855.8 Kg
			Comp. All. Ope. Stress		129.6 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (25.1 \cdot 1454) / (138 \cdot 1 - 0.6 \cdot 25.1) \\
 &= 26.7019 + 1.0000 = 27.7019 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of 0.065 bars} \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 29) / (1454 + 0.6 \cdot 29) \\
 &= 27.2 - 0.065 = 27.1 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 30) / (1453 + 0.6 \cdot 30) \\
 &= 28.1 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (25.1 \cdot (1454 + 0.6 \cdot 29)) / (1 \cdot 29) \\
 &= 127.183 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.022 %

Minimum Design Metal Temperature Results:



Govern. thk, $t_g = 30$, $t_r = 13.4$, $c = 1$ mm., $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.46$, Temp. Reduction = 38 °C

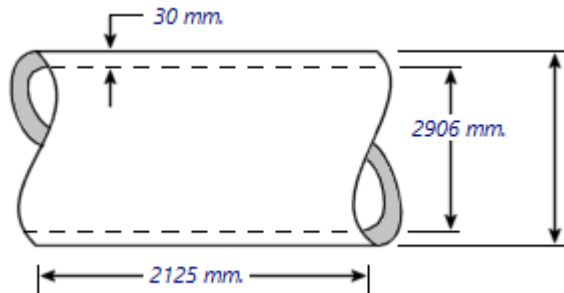
Min Metal Temp. w/o impact per UCS-66, Curve D -30 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	6.000 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	30.000 mm.	Basic MDMT	-30 °C
Stress/Thk. Ratio	0.462	Temperature Difference	18 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E^*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Unclad Shell

Cylindrical Shell Design Information:			
Design Pressure P	25.007 bars	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	0 °C
Length	2125.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm.	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2906.000 mm.	Actual Stress	126.9 N./mm ²
Spec. Min. Thk t _e , or sa	30.0000 mm.	Spec. Nominal Thk.	30.000 mm.
Surface Area	198006.7 cm ²	SG of Contents	0.000
Element Volume	14096710.0 cm ³	Weight of Contents	0.0 Kg
Empty Weight	4557.2 Kg	Operating Weight	4557.2 Kg
		Comp. All. Ope. Stress	129.6 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (25 \cdot 1454) / (138 \cdot 1 - 0.6 \cdot 25) \\
 &= 26.6391 + 1.0000 = 27.6391 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 bars

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 29) / (1454 + 0.6 \cdot 29) \\
 &= 27.2 - 0.007 = 27.2 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 30) / (1453 + 0.6 \cdot 30) \\
 &= 28.1 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (25 \cdot (1454 + 0.6 \cdot 29)) / (1 \cdot 29) \\
 &= 126.888 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.022 %

Minimum Design Metal Temperature Results:



Govern. thk, $t_g = 30$, $t_r = 13.4$, $c = 1$ mm., $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.46$, Temp. Reduction = 38 °C

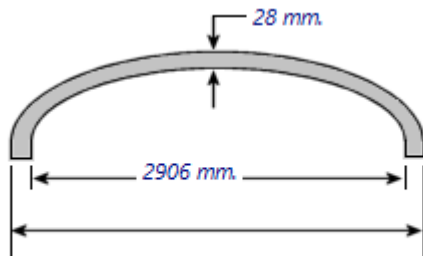
Min Metal Temp. w/o impact per UCS-66, Curve D -30 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	6.000 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	30.000 mm.	Basic MDMT	-30 °C
Stress/Thk. Ratio	0.462	Temperature Difference	18 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E^*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Unclad Topdish

Elliptical Head Design Information:			
Design Pressure P	25.007 bars	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	0 °C
Length	50.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm.	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2906.000 mm.	Actual Stress	134.8 N./mm ²
Spec. Min. Thk t _e , or sa	28.0000 mm.	Spec. Nominal Thk.	34.000 mm.
Surface Area	100546.5 cm ²	SG of Contents	0.000
Element Volume	3544632.0 cm ³	Weight of Contents	0.0 Kg
Empty Weight	2847.7 Kg	Operating Weight	2847.7 Kg
		Comp. All. Ope. Stress	127.9 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 36.325 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.162 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (25 \cdot 2908 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 25)$$

$$= 26.3731 + 1.0000 = 27.3731 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 27) / (1 \cdot 2908 + 0.2 \cdot 27)$$

$$= 25.6 - 0.007 = 25.6 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 28) / (1 \cdot 2906 + 0.2 \cdot 28)$$

$$= 26.5 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (25 \cdot (1 \cdot 2908 + 0.2 \cdot 27)) / (2 \cdot 1 \cdot 27)$$

$$= 134.802 \text{ N./mm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (25 \cdot 1454) / (138 \cdot 1 - 0.6 \cdot 25) + 1$
 $= 27.639 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
Less Operating Hydrostatic Head Pressure of 0.007 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 1 \cdot 33) / (1454 + 0.6 \cdot 33)$
 $= 30.9 - 0.007 = 30.9 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$
 $= (2 + (2908 / (2 \cdot 727)))^2 / 6$
 $= 0.999084$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.990 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 28$, $t_r = 6.32$, $c = 1 \text{ mm.}$, $E^* = 1$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.23$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -32 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 34$, $t_r = 6.34$, $c = 1 \text{ mm.}$, $E^* = 1$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.19$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -28 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	6.000 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	28.000 mm.	Basic MDMT	-32 °C
Stress/Thk. Ratio	0.234	Temperature Difference	72 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 36.325 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.162 mm.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7.800 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	32.500 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	34.120 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	6.600 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	35.750 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	35.750 bars
App 27-4	= M.A.W.P.	6.000 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Clad Dis	178.9	234.0	0.764	34.40
Clad shell 1	168.7	234.0	0.721	34.40
Unclad Shell	168.7	234.0	0.721	34.40
Unclad Topdish	178.9	234.0	0.764	34.40

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1-2"	Nozzle	115.00	115.00	1.000
K1-2"	Nozzle	115.00	101.29	1.135
K3-2"	Nozzle	115.00	101.29	1.135
K5-2"	Nozzle	115.00	101.29	1.135
K7-2"	Nozzle	115.00	101.29	1.135
K2-2"	Nozzle	115.00	101.29	1.135
K4-2"	Nozzle	115.00	101.29	1.135
K6-2"	Nozzle	115.00	101.29	1.135
K8-2"	Nozzle	115.00	101.29	1.135
K10-2"	Nozzle	115.00	101.29	1.135
N1-24"	Nozzle	138.00	138.00	1.000
N1-24"	Pad	138.00	138.00	1.000
N2-24"	Nozzle	138.00	138.00	1.000
N2-24"	Pad	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K9-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Clad Dis	138.00	138.00	1.000
Clad shell 1	138.00	138.00	1.000
Unclad Shell	138.00	138.00	1.000
Unclad Topdish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
D1-2"	20.04	153.00	0.131
K1-2"	7.28	153.00	0.048
K3-2"	7.28	153.00	0.048



K5-2"	7.28	153.00	0.048
K7-2"	7.28	153.00	0.048
K2-2"	7.28	153.00	0.048
K4-2"	7.28	153.00	0.048
K6-2"	7.28	153.00	0.048
K8-2"	7.28	153.00	0.048
K10-2"	7.28	153.00	0.048
N1-24"	73.53	234.00	0.314
N2-24"	64.17	234.00	0.274
M1-24"	64.17	234.00	0.274
K9-2"	7.28	225.00	0.032

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7.800 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	32.500 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	34.378 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	6.600 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	35.750 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	35.750 bars
App 27-4	= M.A.W.P.	6.000 bars

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 25 • 1
 = 32.500 bars

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Clad Dis	177.9	234.0	0.760	33.00
Clad shell 1	167.0	234.0	0.714	32.92
Unclad Shell	166.6	234.0	0.712	32.83
Unclad Topdish	175.9	234.0	0.752	32.63

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1-2"	Nozzle	115.00	115.00	1.000
K1-2"	Nozzle	115.00	101.29	1.135
K3-2"	Nozzle	115.00	101.29	1.135
K5-2"	Nozzle	115.00	101.29	1.135
K7-2"	Nozzle	115.00	101.29	1.135
K2-2"	Nozzle	115.00	101.29	1.135
K4-2"	Nozzle	115.00	101.29	1.135
K6-2"	Nozzle	115.00	101.29	1.135
K8-2"	Nozzle	115.00	101.29	1.135
K10-2"	Nozzle	115.00	101.29	1.135
N1-24"	Nozzle	138.00	138.00	1.000
N1-24"	Pad	138.00	138.00	1.000
N2-24"	Nozzle	138.00	138.00	1.000
N2-24"	Pad	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K9-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Clad Dis	138.00	138.00	1.000
Clad shell 1	138.00	138.00	1.000
Unclad Shell	138.00	138.00	1.000
Unclad Topdish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):



Description	Ambient	Operating	Ratio
D1-2"	19.22	153.00	0.126
K1-2"	6.96	153.00	0.046
K3-2"	6.96	153.00	0.046
K5-2"	6.96	153.00	0.046
K7-2"	6.96	153.00	0.046
K2-2"	6.96	153.00	0.046
K4-2"	6.96	153.00	0.046
K6-2"	6.96	153.00	0.046
K8-2"	6.96	153.00	0.046
K10-2"	6.96	153.00	0.046
N1-24"	70.17	234.00	0.300
N2-24"	61.24	234.00	0.262
M1-24"	61.24	234.00	0.262
K9-2"	6.95	225.00	0.031

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	2700	...	...	No Calc	No Calc
20	30	2700	...	...	No Calc	No Calc
30	40	No Calc	2962	27	0.001266	92.2218
40	50	3609.33	2966	29	0.0010551	86.9314
50	60	3609.33	2966	29	0.0010551	86.9314
60	70	No Calc	2962	27	0.001266	92.2218

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	28	No Calc	...	9.34
40	50	30	No Calc	...	11.3
50	60	30	No Calc	...	11.3
60	70	28	No Calc	...	9.34

Minimum

9.34

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	2700	No Calc	No Calc	No Calc
20	30	2700	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	3609.33	No Calc	No Calc	No Calc
50	60	3609.33	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	2515.11	...	2393.98	...	176.058
20	30	715.122	...	643.831	...	50.0586
30	40	2847.66	3544632	2763.91	3551727	199.336
40	50	1930.09	5970372	1866.39	5978593	135.106
50	60	4557.16	14096710	4406.74	14116122	319.001
60	70	2847.66	3544632	2763.91	3551727	199.336
Total		15412	27156346.00	14838	27198170.00	1078

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
30	Plat	4091.29	...	...	0.20985E-04	PLAT:[1 OF 1]
30	Liqd	3489.51	...	-363.25	...	Liquid: 10
30	Lini	253.313	...	-338.25	...	clad
30	Nozl	6.83601	...	-837.506	...	D1-2"
40	Liqd	3925.76	...	300	...	Liquid: 20
40	Lini	197.218	...	450	...	clad
40	Nozl	11.9662	1556.19	800	...	K1-2"
40	Nozl	11.9662	1532.55	800	-270.229	K3-2"
40	Nozl	11.9662	1532.55	800	270.229	K5-2"
40	Nozl	11.9662	-1556.19	800	...	K7-2"
40	Nozl	11.9662	1532.55	100	-270.229	K2-2"
40	Nozl	11.9662	1556.19	100	...	K4-2"
40	Nozl	11.9662	1532.55	100	270.229	K6-2"
40	Nozl	11.9662	1462.34	100	532.247	K8-2"
40	Nozl	11.9662	-1556.19	100	...	K10-2"
40	Wght	20	...	0.45720E-04	...	nozzles lining
50	Lini	465.655	...	1062.5	...	clad
50	Nozl	523.526	1603	800	...	N1-24"
50	Nozl	535.223	-1603	1400	...	N2-24"
50	Nozl	907.305	...	550	1603	M1-24"
50	Nozl	11.553	1462.34	1550	532.247	K9-2"
50	Wght	130	...	-0.0001463	...	Donut Ring
50	Wght	150	...	-0.0001463	...	Demister pad
50	Wght	850	1000	1000	...	N1 Distributor
50	Wght	700	-1000	1400	...	N2-Hopper
50	Wght	50	...	550	1500	Davit
60	Plat	4700.28	...	837	...	PLAT:[1 OF 1]
60	Lini	253.313	...	388.25	...	clad
60	Wght	360	...	...	...	Clip weight

Total Weight of Each Detail Type:

Platforms	8791.6
Liquid	7415.3
Cladding & Overlay	1169.5
Nozzles	2092.1
Weights	2260.0
Sum of the Detail Weights	21728.5 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	16491.7	16491.7	16491.7	16491.7	16491.7	16491.7
Cladding	1169.5	1169.5	1169.5	1169.5	1169.5	1169.5



Nozzles	2092.1	2092.1	2092.1	2092.1	2092.1	2092.1
Wld Weights	2110.0	2110.0	2110.0	2110.0	2110.0	2110.0
Platforms	...	...	...	8791.6	8791.6	8791.6
Empty Weights	...	...	...	150.0	150.0	...
Test Weights	...	150.0	...	...	...	...
Ope Weights	...	...	...	...	...	150.0
Ope. Liquid	...	...	...	...	...	7415.3
Test Liquid	...	27139.8	...	...	...	...
Totals	21863.3	49153.1	21863.3	30804.9	30804.9	38220.2

Field Installation Options:

Platforms installed before lifting.

* Lining installed after lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	21863.3 kg.
Shop Test Wt.	- Fabricated weight + water (full)	49153.1 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	21863.3 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	30804.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	30804.9 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	38220.2 kg.
Field Test Wt.	- Empty weight + test medium (full)	57798.7 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	16460.6 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	28453.2 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	16693.5 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	157497	156216
20	30	92582.7	91954.4
30	40	100547	96105.5
40	50	83861.7	82165.2
50	60	198007	194001
60	70	100547	96105.5

Total	733041.7	716547.5 cm ²
Total	789.0	771.3 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	2691.17	2691.17	2691.17	...	2691.17
20	30	765.181	765.181	765.181	...	765.181
30	40	3053.84	10887.9	10940.9	0.85859E-04	7398.44
40	50	2192.89	6315.87	8356.84	6.3691	2390.11
50	60	8733.77	9199.42	23287.5	1535.59	9199.42
60	70	3407	8360.6	11903.1	...	8360.6

Cumulative Vessel Weight



From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	30804.9	38220.2	57944.7
20	30	28113.7	35529	55253.5
30	40	27348.6	34763.8	54488.3
40	50	19950.1	23875.9	43547.4
50	60	17560	17560	35190.6
60	70	8360.6	8360.6	11903.1

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	1541.96	1541.96	1541.96
20	30	1541.96	1541.96	1541.96
30	40	1541.96	1541.96	1541.96
40	50	1541.96	1541.96	1541.96
50	60	1535.59	1535.59	1535.59
60	70	...	...	...

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
D1-2"	73.86	82.30	82	600	GR 2.3	53.407	57.387	73.862	57.387
K1-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K3-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K5-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K7-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K2-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K4-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K6-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K8-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K10-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
N1-24"	47.82	51.00	82	300	GR 1.1	12.782	47.817	47.817	47.817
N2-24"	47.82	51.00	82	300	GR 1.1	14.109	47.817	47.817	47.817
M1-24"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K9-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
Min Rating	36.981	41.100, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
D1-2"	77.851	0.50	50	0	6.000	57.387	Pass
N1-24"	657.225	0.50	7266	0	6.000	47.817	Pass
N2-24"	657.225	0.50	7266	4589	6.000	47.817	Pass

Evaluating Flange on Nozzle [D1-2"]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$
 $= 16(50) + 4 \cdot \text{abs}(0)77.851 \leq \pi \cdot 77.851^3 [(73.862 - 6.000) + .500 \cdot 73.862]$
 $= 0.80732E+03 \leq 0.15841E+04$ Check Passes

Maximum Pressure for Flange on Nozzle D1-2" [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(73.862, 0.500 \cdot 73.862 + 73.862 - 807.3 / (\pi \cdot 77.851^3))$
 $= \min(73.862, 57.387)$
 $= 57.387$ bars

Evaluating Flange on Nozzle [N1-24"]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$
 $= 16(7266) + 4 \cdot \text{abs}(0)657.225 \leq \pi \cdot 657.225^3 [(47.817 - 6.000) + .500 \cdot 47.817]$
 $= 0.11625E+06 \leq 0.59776E+06$ Check Passes

Maximum Pressure for Flange on Nozzle N1-24" [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(47.817, 0.500 \cdot 47.817 + 47.817 - 116253.5 / (\pi \cdot 657.225^3))$
 $= \min(47.817, 58.943)$
 $= 47.817$ bars

Evaluating Flange on Nozzle [N2-24"]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$



$$= 16(7266) + 4 \cdot \text{abs}(4589) 657.225 \leq \pi \cdot 657.225^3 [(47.817 - 6.000) + .500 \cdot 47.817]$$
$$= 0.12832\text{E}+06 \leq 0.59776\text{E}+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-24" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$
$$= \min(47.817, 0.500 \cdot 47.817 + 47.817 - 128317.9 / (\pi \cdot 657.225^3))$$
$$= \min(47.817, 57.616)$$
$$= 47.817 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 19.7366 Hz.

The Natural Frequency for the Vessel (Ope...) is 18.7071 Hz.

The Natural Frequency for the Vessel (Filled) is 14.3276 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.95) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$
 $= \min(0.85, 0.9)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	18.707 Hz
Natural Frequency of Vessel (Empty)	19.737 Hz
Natural Frequency of Vessel (Test)	14.328 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.522
Structure Height to Diameter ratio	2.298
Height to top of Structure	6579.500 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm.}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 850 + 300 = 1150 \text{ mm.}$
 $= 3.77 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (3.77 ft.) $< 15 \text{ ft.}$
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900)^{2/9.5}$
 $= 0.85$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1, \text{ per Table 26.6-1}$



As there is No Hill Present: [Kzt]:

$$K1 = 0, K2 = 0, K3 = 0$$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.85 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$$

$$= 16 \text{ psf } [78.1] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16 \cdot 0.85 \cdot 0.52 \cdot 70.7$$

$$= 502 \text{ lbs. } [227] \text{ Kgf}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m2
Skirt2	1150.0	0.000	0.000	0.000	0.849	1.000	78.120
Skirt1	2500.0	0.000	0.000	0.000	0.849	1.000	78.120
Bottom Clad Dis	3025.0	0.000	0.000	0.000	0.849	1.000	78.120
Clad shell 1	3500.0	0.000	0.000	0.000	0.849	1.000	78.120
Unclad Shell	5012.5	0.000	0.000	0.000	0.865	1.000	78.120
Unclad Topdish	6418.4	0.000	0.000	0.000	0.912	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm.
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm.
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(104)$
- L - Total height of structure mm.
- Lc - Total length of conical section(s) of vessel, mm.
- tb - Uncorroded plate thickness at bottom of vessel, mm.
- V30 - Design Wind Speed provided by user, Km/hr
- Vc - Critical wind velocity, Km/hr
- Vw - Maximum wind speed at top of structure, Km/hr
- W - Total corroded weight of structure, Kgf
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kgf
- Z - Maximum amplitude of vibration at top of vessel, mm.
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.32037E-06 (High); 0.32037E-06 < 0.40047E-06 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: 0/5825 = 0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[9.7/19.1^2] \cdot 10^4 = 266$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.40047E-06$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 37606 / (5825 \cdot 2966^2)$$

$$= 0.73390E-06$$

Since Vp is > 0.40047E-06 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[1 OF 1]	30000.00	3000.00	78.12	234.36	1.00
PLAT:[1 OF 1]	65443.20	6912.00	78.12	511.24	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
nozzles linin	0.00	3050.00	78.12	0.00
Donut Ring	0.00	3950.00	78.12	0.00
Demister pad	0.00	3950.00	78.12	0.00
N1 Distributor	0.00	4950.00	78.12	0.00
N2-Hopper	0.00	5350.00	78.12	0.00
Davit	0.00	4500.00	78.12	0.00
Clip weight	0.00	6075.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 18.7071 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1150	3863.19	65674.2	78.12	250.23
20	30	2500	3860.57	38605.7	78.12	147.094
30	40	3025	3880.22	1940.11	78.12	263.058
40	50	3500	3885.46	34969.1	78.12	133.238
50	60	5012.5	3885.46	82566	78.12	314.59
60	70	6418.36	3880.22	24933.6	78.12	700.621

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.050
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(21.50.75)$
 $= 0.200$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/18.7)$
 $= 0.053$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.2, 0.053)$ per 12.8.2
 $= 0.053$

Loads on Building/Structure Supported Equipment:

$$\begin{aligned} &\text{Seismic Force per equation 13.3-1, [Fp]:} \\ &= 0.4 \cdot Ap \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / Ie) \\ &= 0.4 \cdot 2.5 \cdot 0.16 \cdot 38220 (1 + 2 \cdot 0.05) / (1.25 / 1.25) \\ &= 6592.218 \text{ Kgf} \end{aligned}$$

Maximum value of Fp per equation 13.3-2:
 $= 1.6 \cdot Sds \cdot I \cdot W$
 $= 1.6 \cdot 0.16 \cdot 1.25 \cdot 38220 = 11986 \text{ Kgf}$

Minimum value of Fp per equation 13.3-3:
 $= 0.3 \cdot 0.16 \cdot 1.25 \cdot 38220$
 $= 2247.347 \text{ Kgf}$

Total Base Shear $V = Fp$, [V]:
 $= 6592.218 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 38220$
 $= 1198.6 \text{ Kgf}$



Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 18.7071 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	850	2691.17	103.501	102.506	1198.59
20	30	2200	765.181	76.1682	75.4355	1114.19
30	40	2725	10887.9	1342.45	372.908	1090.19
40	50	3200	6315.87	914.471	314.453	748.748
50	60	4712.5	9199.42	1961.55	1844.34	550.682
60	70	5800	8360.6	2194.08	885.501	262.188

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	850	1808.83	6592.22	7448.8	29508.6
20	30	2200	1558.6	6488.72	4586.43	18389.5
30	40	2725	1411.51	6412.55	3101.34	11938.8
40	50	3200	1148.45	5070.1	3037.34	11651.7
50	60	4712.5	1015.21	4155.62	2063.68	7500.06
60	70	5800	700.621	2194.08	240.568	753.368

Note:
 The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	1808.83	0.0010789	0.0040136	0.43731E-05
20	30	1558.6	0.0064818	0.0094483	0.63676E-05
30	40	1411.51	0.0096077	0.0097674	0.63957E-05
40	50	1148.45	0.012695	0.01571	0.67858E-05
50	60	1015.21	0.023086	0.030669	0.71832E-05
60	70	700.621	0.030849	0.031029	0.71833E-05

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	1297.34	8108.39
20	30	1296.46	8102.89
30	40	1303.06	8144.13
40	50	1304.82	8155.13
50	60	1304.82	8155.13
60	70	1303.06	8144.13

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 29.125 Actual deflection : 0.031 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	1107.23	1015.3	80968824	74296768
20	30	922.683	830.694	67518856	60828732
30	40	2580.88	2489.55	187535552	181021472
40	50	2767.12	2675.79	201072960	194567664
50	60	2767.12	2675.79	201072960	194567664
60	70	2580.88	2489.55	187535552	181021472

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	95.7583	188.997	-99.7475	-99.7475
20	30	82.7985	140.397	-91.598	-91.598
30	40	165.6	280.8	-127.912	-127.912
40	50	165.6	280.8	-129.64	-129.64
50	60	165.6	280.8	-129.64	-129.64
60	70	165.6	280.8	-127.912	-127.912

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	66.8374	...	86.8643
40	50	62.1934	...	80.8288
50	60	62.1934	...	80.8288
60	70	66.8374	...	86.8643

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-2.97546	-3.6917	-5.5969	0.20353	0.20353
20	30	-3.31899	-4.19441	-6.52301	0.24859	0.24859
30	40	-1.07731	-1.07731	-1.07731	0.083534	0.083534
40	50	-0.73118	-0.73118	-0.73118	0.077718	0.077718
50	60	-0.64358	-0.64358	-0.64358	0.077397	0.077397
60	70	-0.32934	-0.32934	-0.32934	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	0.20353	0.98319	3.89492	0.49159	...
20	30	0.24859	0.73941	2.96471	0.36971	...
30	40	0.083534	0.16801	0.64677	0.084006	...
40	50	0.077718	0.15309	0.58727	0.076544	...
50	60	0.077397	0.10401	0.37802	0.052007	...
60	70	...	0.013033	0.040813	0.0065163	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	2.16272
20	30	...	...	...	1.67027
30	40	...	...	...	0.37409
40	50	...	...	...	0.3399
50	60	...	...	...	0.20894
60	70	...	...	...	0.016472

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind	Longitudinal Stress Y Forces Seismic
------	----	--------------------------------------	---

		N./mm ²	N./mm ²
10	20	...	0.11577
20	30	...	0.13154
30	40	...	0.042945
40	50	...	0.027442
50	60	...	0.020183
60	70	...	0.010328

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.16	99.75		0.0417
20		82.80	-4.31	91.60		0.0470
30		165.60	-1.33	127.91		0.0104
40		165.60	-0.96	129.64		0.0074
50		165.60	-0.82	129.64		0.0064
60		165.60	-0.34	127.91		0.0027

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-5.23	99.75		0.0524
20		82.80	-5.11	91.60		0.0557
30		165.60	-1.49	127.91		0.0117
40		165.60	-1.12	129.64		0.0086
50		165.60	-0.91	129.64		0.0070
60		165.60	-0.34	127.91		0.0026

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.88	99.75		0.0489



20	82.80	-5.18	91.60	0.0566
30	165.60	-1.33	127.91	0.0104
40	165.60	-0.96	129.64	0.0074
50	165.60	-0.82	129.64	0.0064
60	165.60	-0.34	127.91	0.0027

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.52	95.76	-7.67	99.75	0.0055	0.0769
20		82.80	-7.28	91.60		0.0794
30		165.60	-1.76	127.91		0.0138
40		165.60	-1.37	129.64		0.0106
50		165.60	-1.08	129.64		0.0083
60		165.60	-0.36	127.91		0.0028

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-6.29	99.75		0.0631
20		140.40	-7.14	91.60		0.0780
30		280.80	-1.24	127.91		0.0097
40		280.80	-0.89	129.64		0.0068
50		280.80	-0.77	129.64		0.0060
60		280.80	-0.34	127.91		0.0026

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-5.80	99.75		0.0582
20		140.40	-6.77	91.60		0.0739
30		280.80	-1.16	127.91		0.0091
40		280.80	-0.81	129.64		0.0062
50		280.80	-0.72	129.64		0.0056
60		280.80	-0.33	127.91		0.0026

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.88	99.75		0.0489
20		82.80	-5.18	91.60		0.0566
30	66.01	165.60		127.91	0.3986	
40	61.69	165.60		129.64	0.3725	
50	61.73	165.60		129.64	0.3728	
60	66.52	165.60		127.91	0.4017	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.52	95.76	-7.67	99.75	0.0055	0.0769
20		82.80	-7.28	91.60		0.0794
30	66.53	165.60		127.91	0.4018	
40	62.15	165.60		129.64	0.3753	
50	62.03	165.60		129.64	0.3745	
60	66.56	165.60		127.91	0.4019	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.88	99.75		0.0489
20		82.80	-5.18	91.60		0.0566
30		165.60	-1.33	127.91		0.0104
40		165.60	-0.96	129.64		0.0074



50	165.60	-0.82	129.64	0.0064
60	165.60	-0.34	127.91	0.0027

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.52	95.76	-7.67	99.75	0.0055	0.0769
20		82.80	-7.28	91.60		0.0794
30		165.60	-1.76	127.91		0.0138
40		165.60	-1.37	129.64		0.0106
50		165.60	-1.08	129.64		0.0083
60		165.60	-0.36	127.91		0.0028

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-6.29	99.75		0.0631
20		140.40	-7.14	91.60		0.0780
30	85.95	280.80		127.91	0.3061	
40	80.25	280.80		129.64	0.2858	
50	80.31	280.80		129.64	0.2860	
60	86.54	280.80		127.91	0.3082	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-5.80	99.75		0.0582
20		140.40	-6.77	91.60		0.0739
30	85.87	280.80		127.91	0.3058	
40	80.18	280.80		129.64	0.2855	
50	80.26	280.80		129.64	0.2858	
60	86.53	280.80		127.91	0.3082	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.18	99.75		0.0319
20		82.80	-3.57	91.60		0.0389
30	65.84	165.60		127.91	0.3976	
40	61.54	165.60		129.64	0.3716	
50	61.63	165.60		129.64	0.3721	
60	66.51	165.60		127.91	0.4016	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.39	99.75		0.0339
20		82.80	-3.78	91.60		0.0412
30	65.80	165.60		127.91	0.3973	
40	61.49	165.60		129.64	0.3713	
50	61.57	165.60		129.64	0.3718	
60	66.52	165.60		127.91	0.4017	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.90	99.75		0.0391
20		82.80	-4.44	91.60		0.0485
30	65.84	165.60		127.91	0.3976	
40	61.54	165.60		129.64	0.3716	
50	61.63	165.60		129.64	0.3721	
60	66.51	165.60		127.91	0.4016	



Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.18	99.75		0.0319
20		82.80	-3.57	91.60		0.0389
30	65.84	165.60		127.91	0.3976	
40	61.54	165.60		129.64	0.3716	
50	61.63	165.60		129.64	0.3721	
60	66.51	165.60		127.91	0.4016	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.90	99.75		0.0391
20		82.80	-4.44	91.60		0.0485
30		165.60	-1.16	127.91		0.0091
40		165.60	-0.81	129.64		0.0062
50		165.60	-0.72	129.64		0.0056
60		165.60	-0.33	127.91		0.0026

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		79.80	-3.78	83.12		0.0455
20		69.00	-4.31	76.33		0.0565
30	65.89	138.00		106.59	0.4774	
40	61.57	138.00		108.03	0.4461	
50	61.65	138.00		108.03	0.4467	
60	66.52	138.00		106.59	0.4820	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		79.80	-3.78	83.12		0.0455
20		69.00	-4.31	76.33		0.0565
30		138.00	-1.12	106.59		0.0105
40		138.00	-0.78	108.03		0.0072
50		138.00	-0.70	108.03		0.0065
60		138.00	-0.32	106.59		0.0030

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.90	99.75		0.0391
20		82.80	-4.44	91.60		0.0485
30	65.84	165.60		127.91	0.3976	
40	61.54	165.60		129.64	0.3716	
50	61.63	165.60		129.64	0.3721	
60	66.51	165.60		127.91	0.4016	

Absolute Maximum of the all of the Stress Ratio's

0.4820

Governing Element: Unclad Topdish

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt2	...	-3.6	0.2	...	...	+ Tens
Skirt2	...	-3.6	-0.2	...	-3.8	- Comp
Skirt1	...	-4.1	0.2	...	...	+ Tens
Skirt1	...	-4.1	-0.2	...	-4.3	- Comp

Bottom Clad Dis	66.8	-1.0	0.1	...	65.9	+ Tens
Bottom Clad Dis	66.8	-1.0	-0.1	...	...	- Comp
Clad shell 1	62.2	-0.7	0.1	...	61.6	+ Tens
Clad shell 1	62.2	-0.7	-0.1	...	...	- Comp
Unclad Shell	62.2	-0.6	0.1	...	61.6	+ Tens
Unclad Shell	62.2	-0.6	-0.1	...	...	- Comp
Unclad Topdish	66.8	-0.3	...	...	66.5	+ Tens
Unclad Topdish	66.8	-0.3	...	...	...	- Comp

**Skirt Support Analysis Per EN 13445-3 Section 16.12:**

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt1

Total vessel (empty) weight	Fg	27574.682 Kg
Head Weight below section 2-2	DeltaFg	7398.436 Kg
Weight of vessel contents (liquid)	Ff	7415.269 Kg
Vessel Gross Weight	Fg + Ff	34989.953 Kg

Skirt Mean Diameter	Dz	2938.000 mm.
Skirt Analysis Thickness (Corroded)	ez	9.000 mm.
Skirt Height	H	1000.000 mm.

Additional Axial Force at section 1-1	F1	0.000 Kg
Moment at section 1-1	M1	11938.784 Kg-m.
Applied Moment - Bottom of Skirt	Mb	18389.547 Kg-m.
Weight at Bottom of Skirt	W	35529.016 Kg

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	2935.000 mm.
Head Corroded Thickness	eb	27.000 mm.

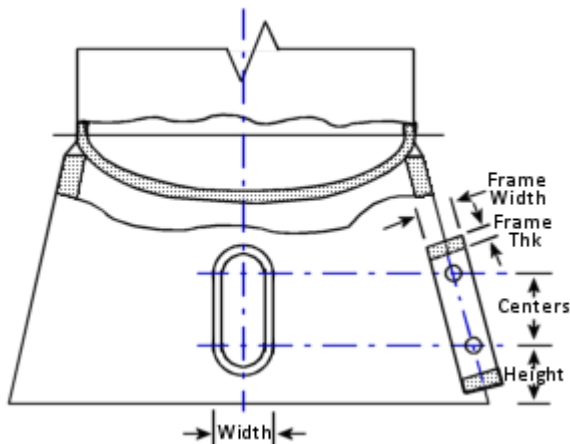
Per EN 13445 Section 7.5.4 [R, r]:

$$h_i = 727 \text{ mm.}$$

$$K = D_i / (2 * h_i) = 2908 / (2 * 727) = 2$$

$$R = D_i (0.44 * K + 0.02) = 2908 * (0.44 * 2 + 0.02) = 2615 \text{ mm.}$$

$$r = D_i ((0.5 / K) - 0.08) = 2908 * ((0.5 / 2) - 0.08) = 495 \text{ mm.}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	8963081.000 cm ⁴
Moment of Inertia	Iyy =	8963081.000 cm ⁴
Section Modulus	W =	61014904.000 mm. ³
Section Metal Area	A =	830.700 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})}$$

$$= \sqrt{(\min(8963081, 8963081) / 831)}$$

$$= 1039 \text{ mm.}$$



Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 1039/1000$$

$$= 1.039$$

Correct the Moment [M4]:

top of skirt is the critical end
 Controlling Moment [M4] = 11939 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -11939/244059616 - 35529/831$$

$$= -4.67 \text{ N./mm}^2$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +11939/244059616 - 35529/831$$

$$= -3.71 \text{ N./mm}^2$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= 0 - 27575 - 7415 + 4 \cdot 11939/2938$$

$$= -18736 \text{ Kg}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= 0 - 27575 - 7415 - 4 \cdot 11939/2938$$

$$= -51244 \text{ Kg}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-18736 + 7398 + 7415) / (\pi \cdot 2935 \cdot 27) + 25 \cdot 2935 / (4 \cdot 27)$$

$$= 67.79 \text{ N./mm}^2$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-51244 + 7398 + 7415) / (\pi \cdot 2935 \cdot 27) + 25 \cdot 2935 / (4 \cdot 27)$$

$$= 66.51 \text{ N./mm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (7415 + 7398) / (\pi \cdot 2935 \cdot 27) + 25 \cdot 2935 / (4 \cdot 27)$$

$$= 68.53 \text{ N./mm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= -18736 / (\pi \cdot 2938 \cdot 9)$$

$$= -2.21 \text{ N./mm}^2$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -51244 / (\pi \cdot 2938 \cdot 9)$$

$$= -6.05 \text{ N./mm}^2$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$



$$= 1 - 0.5(2935 + 27 - 2938 + 9)/(495 + 27)$$

$$= 0.97$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.97) = 14.4 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(27^2 + 9^2 + 2 \cdot 27 \cdot 9 \cdot 0.97)}$$

$$= 17.9 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 17.89 \cdot -18736.0 = -335.25 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 17.89 \cdot -51243.9 = -916.92 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057(eb/ez)^2$$

$$= 0.63 - 0.057(27/9)^2$$

$$= 0.117$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.12 \cdot 6 \cdot -335 / (\pi \cdot 2935 \cdot 27^2)$$

$$= -0.34 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.12 \cdot 6 \cdot -917 / (\pi \cdot 2935 \cdot 27^2)$$

$$= -0.94 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.12 \cdot 6 \cdot -335 / (\pi \cdot 2938 \cdot 9^2)$$

$$= -3.09 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.12 \cdot 6 \cdot -917 / (\pi \cdot 2938 \cdot 9^2)$$

$$= -8.44 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 1.06/116 = 0.0092$$

$$y = 125 \cdot eb/D_b = 125 \cdot 27/2935 = 1.15$$

As this an elliptical head, it will be treated a a Korbbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 1.15 = 3.97$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b ((\gamma / \gamma_A) \cdot \alpha - 1) / (4 \cdot eb)$$

$$= (25 + 0) \cdot 2935 ((14.4/40) \cdot 3.97 - 1) / (4 \cdot 27)$$

$$= 29.47 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 67.8 - 0.34 + 29.5$$

$$= 97.61 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 67.8 + 0.34 - 29.5$$

$$= 37.97 \text{ N./mm}^2$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 66.5 - 0.94 + 29.5$$

$$= 96.92 \text{ N./mm}^2$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 66.5 + 0.94 - 29.5$$

$$= 36.10 \text{ N./mm}^2$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 68.5 + 0.34 + 29.5$$

$$= 97.66 \text{ N./mm}^2$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 68.5 - 0.34 - 29.5$$

$$= 39.40 \text{ N./mm}^2$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 68.5 + 0.94 + 29.5$$

$$= 97.06 \text{ N./mm}^2$$

Stress [σ_{2qotot}]:

$$= \sigma_{2qm} - \sigma_{2qb(o)} - \sigma_{2b(p)}$$

$$= 68.5 - 0.94 - 29.5$$

$$= 39.99 \text{ N./mm}^2$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= -2.21 - 3.09$$

$$= 0.88 \text{ N./mm}^2$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= -2.21 + 3.09$$

$$= -5.30 \text{ N./mm}^2$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -6.05 - 8.44$$

$$= 2.39 \text{ N./mm}^2$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -6.05 + 8.44$$

$$= -14.49 \text{ N./mm}^2$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 138 (3 - (1/1.5)) \cdot (67.8/138)^2$$

$$= 391.80 \text{ N./mm}^2$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 138 (3 - (1/1.5)) \cdot (66.5/138)^2$$

$$= 392.63 \text{ N./mm}^2$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 138 (3 - (1/1.5)) \cdot (68.5/138)^2$$

$$= 391.31 \text{ N./mm}^2$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$



$$= 138 (3 - (1/1.5) \cdot (68.5/138)^2)$$

$$= 391.31 \text{ N./mm}^2$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 138 (3 - (1/1.5) \cdot (-2.21/138)^2)$$

$$= 413.98 \text{ N./mm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 138 (3 - (1/1.5) \cdot (-6.05/138)^2)$$

$$= 413.82 \text{ N./mm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm2)

Stress	Actual	Allowed	Result	Equation
Stot1pi	97.61	391.80	Pass	46
Stot1po	37.97	391.80	Pass	47
Stot1qi	96.92	392.63	Pass	48
Stot1qo	36.10	392.63	Pass	49
Stot2pi	97.66	391.31	Pass	50
Stot2po	39.40	391.31	Pass	51
Stot2qi	97.06	391.31	Pass	52
Stot2qo	39.99	391.31	Pass	53
Stot3pi	0.88	413.98	Pass	54
Stot3po	-5.30	413.98	Pass	55
Stot3qi	2.39	413.82	Pass	56
Stot3qo	-14.49	413.82	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.
 Lining is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	4791.493 mm.
Center of Gravity of the Liquid	2714.356 mm.
Center of Gravity of the Lining	4262.500 mm.
Center of Gravity of the Nozzles	4424.416 mm.
Center of Gravity of the Added Weights (Operating)	4802.433 mm.
Center of Gravity of the Added Weights (Empty)	4802.433 mm.
Center of Gravity of Bare Shell New and Cold	3609.942 mm.
Center of Gravity of Bare Shell Corroded	3622.462 mm.
Vessel CG in the Operating Condition	3852.128 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	3820.989 mm.
Vessel CG in the Test Condition	4070.709 mm.

Rigging Analysis Results:

• Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	5825.00 mm.
Total vessel weight (No Liquid) Twt	30804.92 Kgf
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	46207.38 Kgf
Elevation of the Tailing Lug (bottom)	150.00 mm.
Elevation of the Lifting Lug (top)	5475.00 mm.
Design Reaction force at the tailing lug	14352.59 Kgf
Design Reaction force at the lifting lug	31854.79 Kgf
CG distance during field lifting operation	3820.99 mm.
CG distance from tailing lug	3670.99 mm.
CG distance from the nearer lifting lug/trunnion	1654.01 mm.

Compressive Allowable Increase Factor 1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	2.90	2500.00	80.08 (UG-23)
Shear	1.10	0.00	79.80 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	0.0	0.0	12412.9	1.1
1700.00	14693.9	2.1	391.2	0.0
2700.00	19037.3	1.0	2990.2	0.1
2750.00	19089.2	0.9	2990.2	0.1
3650.00	17835.8	0.9	7851.4	0.3
5775.00	2394.6	0.1	1007.0	0.0

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.036 at elevation 2500 mm. - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.

Process Air 3rd KOD 198-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 3rd KO Drum-198-D-7% Page 63 of 145

Center of Gravity Calculation: Step: 23 2:30pm Sep 16,2026



HEXAGON

--- Plot data successfully generated ...----

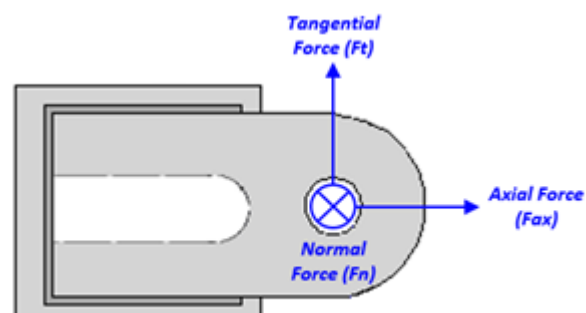
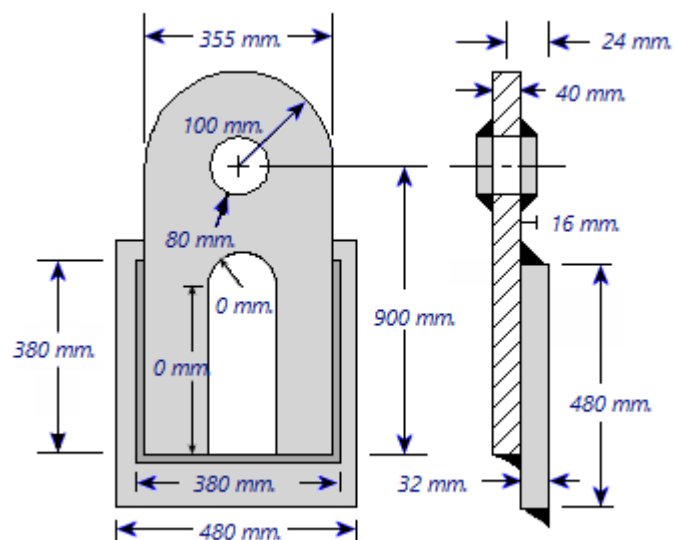
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Lifting Lug Calculations:

Input Values:

Lifting Lug Material		SA-516 70
Lifting Lug Yield Stress	Yield	262.00 N./mm ²
Width of Lifting Lug	w	355.0000 mm.
Thickness of Lifting Lug	t	40.0000 mm.
Diameter of Hole in Lifting Lug	dh	80.0000 mm.
Radius of Semi-Circular Arc of Lifting Lug	r	100.0000 mm.
Height of Lug from Bottom to Center of Hole	h	900.0000 mm.
Offset from Vessel OD to Center of Hole	off	24.0000 mm.
Lug Fillet Weld Size	tw	16.0000 mm.
Length of weld along side of Lifting Lug	wl	380.0000 mm.
Length of Weld along Bottom of Lifting Lug	wb	380.0000 mm.
Thickness of Collar (if any)	tc	0.0000 mm.
Diameter of Collar (if any)	dc	0.0000 mm.
Impact Factor	Impfac	1.50
Number of Lugs in Group		2
Reinforcing Pad Circumferential Width	wpb	480.00 mm.
Reinforcing Pad Longitudinal Length	wpl	480.00 mm.
Reinforcing Pad Thickness	thkp	32.00 mm.
Reinforcing Pad Fillet Weld Size	tfp	16.00 mm.

Lifting Lug Orientation to Vessel: Flat
 Lift Orientation : From Horizontal to Vertical



PV Elite computes the forces and stresses from 0 to 90 degrees at 1 degree increments. The table below provides the results in 5 degree increments.

The 0 degree angle position is in the horizontal direction and the 90 degree angle position is in the vertical direction.

PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

Lifting Lug and Weld Stresses at various Angles: N./mm²

Angle	----- Weld Stresses			----- Lug Stresses		Maximum
	Primary	Secondary	Allowable	Combined	Allowable	Ratio
0	14.2	83.6	104.8	96.7	172.9	0.798
5	14.7	84.3	104.8	98.4	172.9	0.805
10	15.3	84.3	104.8	99.4	172.9	0.804
15	15.8	83.6	104.8	99.6	172.9	0.798
20	16.3	82.2	104.8	99.0	172.9	0.784
25	16.8	80.1	104.8	97.5	172.9	0.765
30	17.3	77.4	104.8	95.4	172.9	0.739
35	17.7	74.0	104.8	92.4	172.9	0.707
40	18.1	70.1	104.8	88.7	172.9	0.668
45	18.5	65.5	104.8	84.3	172.9	0.625
50	18.9	60.3	104.8	79.2	172.9	0.576
55	19.3	54.6	104.8	73.4	172.9	0.521
60	19.7	48.4	104.8	67.0	172.9	0.462
65	20.1	41.7	104.8	60.0	172.9	0.398
70	20.6	34.6	104.8	52.4	172.9	0.330
75	21.1	26.9	104.8	44.3	172.9	0.257
80	21.8	18.7	104.8	35.6	172.9	0.208
85	22.7	9.9	104.8	26.2	172.9	0.216
90	24.1	2.1	104.8	16.0	172.9	0.230

Computed Results:

Total vessel weight (no liquid)	30804.92 Kg
Design reaction force at the tailing lug	14352.59 Kg
Design reaction force at the lifting lug	31854.79 Kg
Worst case lift angle for primary weld stresses	90 deg
Tangential force at this angle	0.0 Kg
Axial force at this angle	23103.7 Kg
Worst case lift angle for secondary weld stresses	7 deg
Tangential force at this angle	16071.6 Kg
Axial force at this angle	1973.3 Kg
Worst case lift angle for lug stresses	14 deg
Tangential force at this angle	15957.2 Kg
Axial force at this angle	3978.6 Kg

Converting the weld leg dimension (tw) to the weld throat dimension.

Weld Group Inertia Calculations:

Weld group inertia about the longitudinal axis	I _{ll}	21320.057 cm ⁴
Weld group centroid distance in the long. direction	Y _{ll}	255.219 mm.
Dist. of weld group centroid from lug bottom	Y _{llb}	124.781 mm.
Weld group inertia about the circumferential axis	I _{lc}	34397.082 cm ⁴
Weld group centroid distance in the circ. direction	Y _{lc}	190.000 mm.

Note: The Impact Factor is applied to the Forces acting on the Lug.

Primary Shear Stress in the Welds due to Shear Loads [S_{sl}]:

$$\begin{aligned}
 &= \sqrt{(F_{ax}^2 + F_t^2 + F_n^2)} / ((2 \cdot w_l + w_b) \cdot t_w) \\
 &= \sqrt{(23104^2 + 0^2 + 0^2)} / ((2 \cdot 380 + 380) \cdot 11.3) \\
 &= 17.57 \text{ N./mm}^2
 \end{aligned}$$



Shear Stress in the Welds due to Bending Loads [Sblf]:

$$= (F_n(h-Y_{lb}))Y_{ll}/I_{ll} + (F_{ax} \cdot \text{off} \cdot Y_{ll}/I_{ll}) + (F_t \cdot \text{off} \cdot Y_{lc}/I_{lc})$$

$$= (0(900-125))255/21320 +$$

$$(23104 \cdot 24 \cdot 255/21320) +$$

$$(0 \cdot 24 \cdot 190/34397)$$

$$= 6.51 \text{ N./mm}^2$$

Total Shear Stress for Combined Loads [St]:

$$= S_{sll} + S_{blf}$$

$$= 17.6 + 6.51$$

$$= 24.08 \text{ N./mm}^2$$

Allowable Shear Stress for Combined Loads [Sta]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac (AISC Shear Allowable)}$$

$$= 0.4 \cdot 262 \cdot 1$$

$$= 104.80 \text{ N./mm}^2$$

Secondary Shear Stress in the Welds due to Shear Loads:

Unit Weld Polar Section Modulus [Uwsm]:

$$= (2 \cdot w_l + w)^3 / 12 - w_l^2(w_l + w)^2 / (2 \cdot w_l + w)$$

$$= 45553568.00 \text{ mm}^3$$

Loads on Welds due to Torsional Moment:

Shear Stress in Normal Direction [Fth]:

$$= F_t \cdot (h - (w_l - \text{Cent})) \cdot (w_b/2) / U_{wsm}$$

$$= 51.65 \text{ Kg/mm.}$$

Shear Stress in Circumferential Direction [Ftv]:

$$= F_t \cdot (h - (w_l - \text{Cent})) \cdot \text{Cent} / U_{wsm}$$

$$= 68.09 \text{ Kg/mm.}$$

Primary Shear Stress [Fsv]:

$$= F_t / (2 \cdot w_l + w)$$

$$= 14.41 \text{ Kg/mm.}$$

Resultant Load on Weld Group [Fr]:

$$= \sqrt{(F_{th}^2 + (F_{tv} + F_{sv})^2)}$$

$$= \sqrt{(51.6^2 + (68.1 + 14.4)^2)}$$

$$= 97.34 \text{ Kg/mm.}$$

Resultant Secondary Weld Stress [Fws]:

$$= F_r / T_{weld}$$

$$= 97.3/11.3$$

$$= 84.39 \text{ N./mm}^2$$

Allowable Resultant Secondary Weld Stress [Psa]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1$$

$$= 104.80 \text{ N./mm}^2$$

Shear Stress in Lug above Hole [Shs]:

$$= \sqrt{(P_l^2 + F_n^2)} / S_{ha}$$

$$= \sqrt{(23104^2 + 0^2)} / 48$$

$$= 47.20 \text{ N./mm}^2$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1$$

$$= 104.80 \text{ N./mm}^2$$

Pin Hole Bearing Width [Pbw]:

$$= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness}$$

$$= 40 + 2 \cdot 0$$

$$= 40.000 \text{ mm.}$$

Pin Hole Bearing Stress [Pbs]:

$$= \sqrt{(P_l^2 + F_n^2)} / (P_{bw} \cdot dh)$$

$$= \sqrt{(23104^2 + 0^2)} / (40 \cdot 80)$$

$$= 70.80 \text{ N./mm}^2$$

Allowable Bearing Stress [Pba]:

$$= \min(0.75 \cdot \text{Yield} \cdot \text{Occfac}, 0.9 \cdot \text{Yield}) \text{ AISC Bearing All.}$$

$$= \min(0.75 \cdot 262 \cdot 1, 236)$$

$$= 196.50 \text{ N./mm}^2$$

Bending Stress at the Base of the Lug [Fbs]:

$$= F_n \cdot (h - w_l) / (w \cdot t^2 / 6) + F_t \cdot (h - w_l) / (w^2 \cdot t / 6)$$

$$= 0 \cdot (900 - 380) / (355 \cdot 40^2 / 6) +$$

$$15957 \cdot (900 - 380) / (355^2 \cdot 40 / 6)$$

$$= 96.86 \text{ N./mm}^2$$

Tensile Stress at the Base of the Lug [Fa]:

$$= F_{ax} / (w \cdot t)$$

$$= 3979 / (355 \cdot 40)$$

$$= 2.75 \text{ N./mm}^2$$

Total Combined Stress at the Base of the Lug:

$$= F_{bs} + F_a$$

$$= 96.9 + 2.75$$

$$= 99.60 \text{ N./mm}^2$$

Lug Allowable Stress for Bending and Tension:

$$= \min(0.66 \cdot \text{Yield} \cdot \text{Occfac}, 0.75 \cdot \text{Yield})$$

$$= \min(0.66 \cdot 262 \cdot 1, 196)$$

$$= 172.92 \text{ N./mm}^2$$

Required Shackle Pin Diameter [Spd]:

$$= \sqrt{((2(P_l^2 + F_n^2))^{1/2} / (\pi \cdot \text{Sta}))}$$

$$= \sqrt{(2(23104^2 + 0^2))^{1/2} / (\pi \cdot 105)}$$

$$= 37.0995 \text{ mm.}$$

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

The full lug load is applied in both the circ. and long. directions simultaneously.

Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.

Input Echo, WRC107/537 Item 1, Description: Lift Lug

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	2906.000 mm.
Vessel Thickness	Tv	30.000 mm.

Note: Lifting lug calculations are performed at ambient temperature conditions.

Attachment Type	Type	Rectangular
Parameter C11	C11	387.00 mm.
Parameter C22	C22	396.00 mm.
Thickness of Reinforcing Pad	Tpad	32.000 mm.
Pad Parameter C11P	C11p	512.000 mm.
Pad Parameter C22P	C22p	496.000 mm.
Design Internal Pressure	Dp	0.000 bars
Include Pressure Thrust		No

External Forces and Moments in WRC 107/537 Convention:

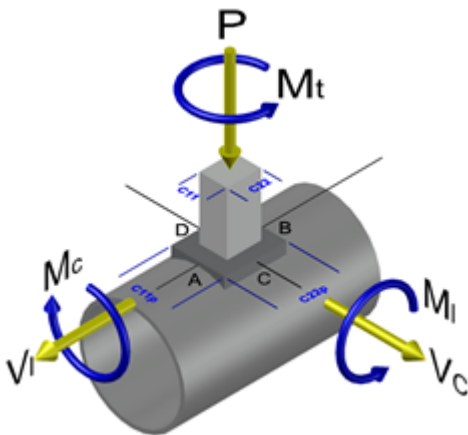
Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	Vl	23103.7	Kgf
Circumferential Shear (SUS)	Vc	23103.7	Kgf
Circumferential Moment (SUS)	Mc	0.0	Kg-m.

Longitudinal Moment (SUS) Ml 0.0 Kg-m.
 Torsional Moment (SUS) Mt 16403.9 Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSstained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	23103.7	Kgf
Longitudinal Shear	VL	23103.7	Kgf
Circumferential Moment	MC	0.0	Kg-m.
Longitudinal Moment	ML	0.0	Kg-m.
Torsional Moment	MT	16403.9	Kg-m.

Dimensionless Parameters used : Gamma = 23.94

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.134	4C	4.108	(A,B)
N(PHI) / (P/Rm)	0.134	3C	3.483	(C,D)
M(PHI) / (P)	0.131	2C1	0.086	(A,B)
M(PHI) / (P)	0.131	1C	0.121	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.131	3A	0.729	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.147	1A	0.094	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.132	3B	2.453	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.136	1B	0.044	(A,B,C,D)
N(x) / (P/Rm)	0.133	3C	3.499	(A,B)
N(x) / (P/Rm)	0.133	4C	4.119	(C,D)
M(x) / (P)	0.133	1C1	0.123	(A,B)
M(x) / (P)	0.133	2C	0.085	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.131	4A	1.104	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.161	2A	0.048	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.132	4B	0.708	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.141	2B	0.071	(A,B,C,D)



Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear VC		4.7	4.7	-4.7	-4.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-4.6	-4.6	4.6	4.6
Shear MT		67.7	67.7	67.7	67.7	67.7	67.7	67.7	67.7
Tot. Shear		72.4	72.4	63.0	63.0	63.1	63.1	72.3	72.3
Str. Int.		144.9	144.9	126.0	126.0	126.2	126.2	144.7	144.7

Dimensionless Parameters used : $\Gamma = 48.93$

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
$N(\Phi) / (P/R_m)$	0.171	4C	7.103	(A,B)
$N(\Phi) / (P/R_m)$	0.171	3C	4.903	(C,D)
$M(\Phi) / (P)$	0.173	2C1	0.038	(A,B)
$M(\Phi) / (P)$	0.173	1C	0.070	(C,D)
$N(\Phi) / (MC/(R_m^{*2} * \beta))$	0.173	3A	1.986	(A,B,C,D)
$M(\Phi) / (MC/(R_m * \beta))$	0.191	1A	0.072	(A,B,C,D)
$N(\Phi) / (ML/(R_m^{*2} * \beta))$	0.171	3B	5.148	(A,B,C,D)
$M(\Phi) / (ML/(R_m * \beta))$	0.176	1B	0.026	(A,B,C,D)
$N(x) / (P/R_m)$	0.173	3C	4.859	(A,B)
$N(x) / (P/R_m)$	0.173	4C	7.078	(C,D)
$M(x) / (P)$	0.172	1C1	0.073	(A,B)
$M(x) / (P)$	0.172	2C	0.038	(C,D)
$N(x) / (MC/(R_m^{*2} * \beta))$	0.173	4A	3.628	(A,B,C,D)
$M(x) / (MC/(R_m * \beta))$	0.205	2A	0.033	(A,B,C,D)
$N(x) / (ML/(R_m^{*2} * \beta))$	0.171	4B	1.960	(A,B,C,D)
$M(x) / (ML/(R_m * \beta))$	0.180	2B	0.036	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0



Circ. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Memb. P	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P	-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear VC	7.4	7.4	-7.4	-7.4	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-7.6	-7.6	7.6	7.6
Shear MT	84.5	84.5	84.5	84.5	84.5	84.5	84.5	84.5
Tot. Shear	91.8	91.8	77.1	77.1	76.8	76.8	92.1	92.1
Str. Int.	183.7	183.7	154.2	154.2	153.7	153.7	184.1	184.1

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		4.7	4.7	-4.7	-4.7	-4.6	-4.6	4.6	4.6
Shear Q (SUS)		67.7	67.7	67.7	67.7	67.7	67.7	67.7	67.7
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		9.4	9.4	9.4	9.4	9.2	9.2	9.2	9.2
Pm+Pl+Q (Total)		144.9	144.9	126.0	126.0	126.2	126.2	144.7	144.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	9.44	207.00	Passed
Pm+Pl+Q (TOTAL)	144.89	414.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:



Vessel Stress Summation at Reinforcing Pad Edge (N/mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		7.4	7.4	-7.4	-7.4	-7.6	-7.6	7.6	7.6
Shear Q (SUS)		84.5	84.5	84.5	84.5	84.5	84.5	84.5	84.5
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		14.8	14.8	14.8	14.8	15.2	15.2	15.2	15.2
Pm+Pl+Q (Total)		183.7	183.7	154.2	154.2	153.7	153.7	184.1	184.1

Vessel Stress Summation Comparison (N/mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	15.23	207.00	Passed
Pm+Pl+Q (TOTAL)	184.15	414.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	2949.0000	mm.
Skirt Thickness	STHK	12.0000	mm.
Skirt Internal Corrosion Allowance	SCA	1.0000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material	SA-36		

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	32.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	N./mm ²
Basing Yield Stress		262.00	N./mm ²
Inside Diameter of Basing	DI, D4	2769.0000	mm.
Outside Diameter of Basing	DOU, D5	3228.9998	mm.
Nominal Diameter of Bolts	BND, d(b0)	42.0000	mm.
Bolt Corrosion Allowance	BCA	0.0000	mm.
Root Area of a Single Bolt	Area	8.6100	cm ²
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	138.00	N./mm ²
Number of Bolts	RN, nb	16	
Diameter of Bolt Circle	DC, DBC	3088.9998	mm.
Bolt Allowable Shear Stress		82.800	N./mm ²
Ultimate Comp. Strength of Concrete	FPC	41.0	N./mm ²
Allowable Comp. Strength of Concrete	FC	22.5	N./mm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm.
Width of Gussets at Top Plate	TWDT	130.0000	mm.
Width of Gussets at Base Plate	BWDT	130.0000	mm.
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	262.0	N./mm ²
Height of Chair	h1	300.0000	mm.
Distance between Gussets	RG, b6	90.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	45.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	38.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	140.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	58.0000	mm.
External Corrosion Allowance	CA	1.0000	mm.
Dead Weight of Vessel	DW	30804.9	Kgf
Operating Weight of Vessel	ROW	38220.2	Kgf
Earthquake Moment on Basing	EQMOM	29658.6	Kg-m.
Wind Moment on Basing	WIMOM	7598.8	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 29659/3089) - 37022) / 16$$



= 86.4 Kgf

Bolt Load, Windload Empty Case:

= $((4 \cdot M/DC) - W) / nb$ per Jawad & Farr, Eq. 12.3

= $((4 \cdot 7599/3089) - 30805) / 16$

= -1310.3 Kgf

Bolt Load, Windload Vortex Shedding Case:

= $((4 \cdot M/DC) - W) / nb$ per Jawad & Farr, Eq. 12.3

= $((4 \cdot 0/3089) - 38220) / 16$

= -2388.8 Kgf

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in one Bolt Abs : 8.6100 cm²

Area Available in all the Bolts Abs * RN : 137.7600 cm²



*There seems to be no tensile load on bolts per the Brownell and Young Method.
It is possible that this method is encountering a problem. It is strongly
recommended that you try the simplified analysis method for the basing.*

Check the Bearing Stress in the Concrete [fc(max)]

= $fc[(2kd + t3) / (2kd)]$

= $0.31[(2 \cdot 1 \cdot 3089 + 230) / (2 \cdot 1 \cdot 3089)]$

= 0.32 N./mm², Should be ≤ 22.5 N./mm²

Values for table 10.3, l = 140, b = 303, l/b = 0.46164685

Maximum Moment per unit width [Mmax]:

= max(Mx, My) = max(73.1, 220) = 220 Kgf

Reqd Thickness of Basing, Brownell & Young Method [T]:

= $\sqrt{(6 \cdot Mmax / fallow)} + Ca$

= $\sqrt{(6 \cdot 220/173)} + 1$

= 9.65 mm.

Nomenclature:

a = (Dc-Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abs	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 140, b = 90, b/l = 0.64285713

As b/l (0.64) is less than 1, inverting b/l = 1.56 0.

Moment Term, based on geometry [Mo]:

= $P / (4 \cdot \pi) [1.3(\ln((2 \sin(\pi \cdot a/l) / (\pi \cdot g))) + 1)] - [(0.7 - g2)P / (4\pi)]$

= $12116 / (4 \cdot \pi) [1.3(\ln((2 \cdot 140 \cdot \sin(\pi \cdot 70/140)) / (\pi \cdot 34.9))) + 1] - [(0.7 - 0.065) \cdot 12116 / (4 \cdot \pi)]$

= 1525.7030 Kgf

Required Thickness of Continuous Top Ring [Tc]:

= $\sqrt{(6 \cdot Abs(Mo) / Fb)} + Ca$

= $\sqrt{(6 \cdot Abs(1526) / 175)} + 1$

= 23.6712 mm.

Required Gusset Plate Thickness [tg]:

= $P / (Stress Term \cdot lavg) + Ca$

= $12116 / (124 \cdot 130) + 1$

= 10.5 (not less than 9.52 + 1) mm.

Bolt spacing [m]:



$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 3089/16 = 607 \text{ mm.}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76(P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot Sktope))^{2/3} \cdot r^{1/3} + Ca$$

$$= 1.76(12116 \cdot 70 / (607 \cdot 262 \cdot 171))^{2/3} \cdot 14741^{1/3} + Ca$$

$$= 9.1 + 1 = 10.1 \text{ mm.}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 6592 / (8.61 \cdot 16)$$

$$= 4.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	9.6534	mm.
Actual Basing Thickness as entered by user	32.0000	mm.
Required Thickness of Chair Cap	23.6712	mm.
Actual Top Ring Thickness as entered by user	38.0000	mm.
Required Gusset thickness, + CA	10.5250	mm.
Actual Gusset Thickness as entered by user	16.0000	mm.
Required Thickness of Skirt for Local Stress	10.0982	mm.
Given Thickness of Skirt	12.0000	mm.
Required Gusset Height to meet local stress	165.0816	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$$

$$= 7.67 \cdot (12 - 1)$$

$$= 8.61 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 8.61 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 0.65 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (Hg + Tta))$$

$$= 12116 / (162 + 2 \cdot (230 + 38))$$

$$= 17.4 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (Hg + Tta) + 0.6667 \cdot (Hg + Tta)^2)$$

$$= 12116 \cdot 70 / (162 \cdot (268) + 0.6667 \cdot (268)^2)$$

$$= 9.29 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{ Wv^2 + Wh^2 }$$

$$= \sqrt{ 17.4^2 + 9.29^2 }$$

$$= 19.7 \text{ Kg/mm.}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 19.7 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 1.48 \text{ mm.}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 12116 / (2 \cdot 130)$$

$$= 46.6 \text{ Kg/mm.}$$

Weld Load [Wh]:
 = Bolt Load • e / (2 • Hgt • TopWth)
 = 12116 • 70 / (2 • 268 • 130)
 = 12.2 Kg/mm.

Resultant Weld Load [Wr]:
 = $\sqrt{(Wv^2 + Wh^2)}$
 = $\sqrt{(46.6^2 + 12.2^2)}$
 = 48.2 Kg/mm.

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
 = Wr / [(0.4 • Yield) • 2 • 0.707]
 = 48.2 / [(0.4 • 230) • 2 • 0.707]
 = 3.63 mm.

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basinger to Skirt Double Fillet Weld Size	4.7625 mm.
Required Gusset to Skirt Double Fillet Weld Size	6.3500 mm.
Required Top Plate to Skirt Weld Size	6.3500 mm.
Required Gusset to Top Plate Double Fillet Weld Size	3.6295 mm.

Tailing Lug Stress Calculation Results :

Tail Lug Type:	Single		
Tail lug yield stress	Syt	262.00	N./mm
Tail lug allowable stress	Sat	138.00	N./mm
Tail lug thickness	TTL	40.0000	mm.
Pin hole diameter	DPH	80.0000	mm.
Fillet weld size	WS	20.0000	mm.
Tail lug load	TLOAD	14352	Kgf

Decay Length Along the Skirt (e):
 = $0.5 \cdot \sqrt{(DS/2 \cdot Ts)}$
 = $0.5 \cdot \sqrt{(2949/2 \cdot 12)}$
 = 66.51 mm.

Skirt Effective Length :
 = HG + TTA + e
 = 230 + 38 + 66.5
 = 334.51 mm.

Section	CG (Y) mm.	Width (H) mm.	Area (A) cm ²	Moment Area mm. ³
Top plate, t	160.00	140.00	53.20	851200.06
Skirt, s	84.00	334.51	40.14	337185.59
Bottom plate, b	115.00	230.00	73.60	846399.25
		TOTAL	166.94	2034784.88

Calculated Overall CG (Y):
 = Mtotal / Atotal
 = 2034784/167
 = 121.89 mm.

Calculated Moment of Inertia (I):
 = (At • Yt² + Topwth³ • TTA/12 + As • Ys² +
 Ts³ • Hs/12 + Ab • Yb² + Hb³ • Tb/12) -
 Mtotal²/Atotal
 = (53.2 • 160² + 140³ • 38/12 + 40.1 • 84² +
 12³ • 335/12 + 73.6 • 115² + 230³ • 32/12) -
 2034784²/167
 = 5502.17 cm⁴



Required Section Modulus (Zr) - per Roark's Table 17 ref 20

$$= \text{TLOAD} \cdot (DI/2 + Y) \cdot (K2 + 0.5) / (0.66 \cdot Sy \cdot 2 \cdot \pi)$$

$$= 14353 \cdot (2769/2 + 122) \cdot (1 + 0.5) / (0.66 \cdot 262 \cdot 2 \cdot 3.14)$$

$$= 292731.38 \text{ mm}^3$$

Calculated Section Modulus (Zc):

$$= I / Y$$

$$= 5502/122$$

$$= 451417.91 \text{ mm}^3$$

Allowable Shear at the Pin Hole (Ssa):

$$= 0.4 \cdot Syt$$

$$= 0.4 \cdot 262$$

$$= 104.80 \text{ N./mm}^2$$

Tail Lug Radius (R):

$$= (TBA + HG + TTA) / 2$$

$$= (32 + 230 + 38) / 2$$

$$= 150.00 \text{ mm}$$

Calculated Shear at the Pin Hole (Ssc):

$$= \text{TLOAD} / (2 \cdot TTL \cdot (R - DPH / 2))$$

$$= 14353 / (2 \cdot 40 \cdot (150 - 80/2))$$

$$= 15.99 \text{ N./mm}$$

Allowable Bearing Stress at the Pin Hole (Sba):

$$= 0.75 \cdot Syt$$

$$= 0.75 \cdot 262$$

$$= 196.50 \text{ N./mm}^2$$

Calculated Bearing Stress at the Pin Hole (Sbc):

$$= \text{TLOAD} / (DPH \cdot TTL)$$

$$= 14353 / (80 \cdot 40)$$

$$= 43.99 \text{ N./mm}^2$$

Weld Length (LW):

$$= 2 \cdot (\text{TOPWTH} + TBA + TTA + HG + (\text{DOU} - \text{DS}) / 2)$$

$$= 2 \cdot (140 + 32 + 38 + 230 + (3229 - 2949) / 2)$$

$$= 1143.00 \text{ mm}$$

Allowable Weld Stress (Swa):

$$= 0.49 \cdot Sat$$

$$= 0.49 \cdot 138$$

$$= 67.62 \text{ N./mm}^2$$

Calculated Weld Stress (Swc):

$$= \text{TLOAD} / (0.707 \cdot WS \cdot LW)$$

$$= 14353 / (0.707 \cdot 20 \cdot 1143)$$

$$= 8.71 \text{ N./mm}^2$$

Tailing Lug Results Summary (mm.³ & N./mm²):

	Actual	Allowable or Required	Pass/Fail
Section Modulus	451417.91	292731.38	Passed
Shear Stress	15.99	104.80	Passed
Bearing Stress	43.99	196.50	Passed
Weld Stress	8.71	67.62	Passed

Review notes about nozzle loadings and supports in the Warnings report.

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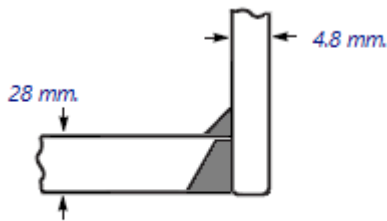
**Input, Nozzle Desc: D1-2"****From: 30**

Pressure for Reinforcement Calculations	P	6.141	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2906.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	28.0000	mm.
Head Internal Corrosion Allowance	cas	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP316L	
Material UNS Number		S31603	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	115.00	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80S	
Flange Material		SA-182 F316L	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	28.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 2.3	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1600.0	mm.
Centerline Distance to Tangent		1300.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: D1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.993 in.
 Actual Thickness Used in Calculation 0.191 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.14 \cdot 2908 \cdot 0.9 / (2 \cdot 138 \cdot 1 \cdot 0.2 \cdot 6.14)$
 $= 5.8221$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.14 \cdot 25.3 / (115 \cdot 1 \cdot 0.6 \cdot 6.14)$
 $= 0.1356$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3250	mm.
Parallel to Vessel Wall	Rn+tn+t	57.1625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	12.1126	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: D1-2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1356	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 7.4673	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.4673	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 3.4200	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[3.42, \max(7.47, 2.5)]$
 $= 3.4200$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.14, 3.42)$
 $= 3.4200$ mm.

Available Nozzle Neck Thickness = 4.8450 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
 (see 319.4.4 and 302.3.5):**

Sustained	:	51.5,	Allowable	:	115.0 N./mm2	Passed
Expansion	:	0.0,	Allowable	:	236.0 N./mm2	Passed
Occasional	:	1.5,	Allowable	:	152.9 N./mm2	Passed



Shear : 34.8, Allowable : 80.5 N./mm2 Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: D1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 4.8450 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	3.3915 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1725 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 178.1725 mm.

Input Echo, WRC107/537 Item 1, Description: D1-2" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm.
Vessel Diameter	Dv	5230.800	mm.
Vessel Thickness	Tv	28.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm2
Vessel Hot S.I. Allowable	Smh	138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	0.0000	mm.
Nozzle Diameter	Dn	50.635	mm.
Nozzle Thickness	Tn	4.845	mm.
Nozzle Material		SA-312 TP316L	
Nozzle UNS Number		S31603	
Nozzle Cold S.I. Allowable	SNmc	115.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	115.00	N./mm2
Design Internal Pressure	Dp	6.141	bars
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	382.4	Kgf
Longitudinal Shear (SUS)	(V1) V1	254.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	254.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	35.7	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	35.7	Kg-m.
Torsional Moment (SUS)	Mt	51.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
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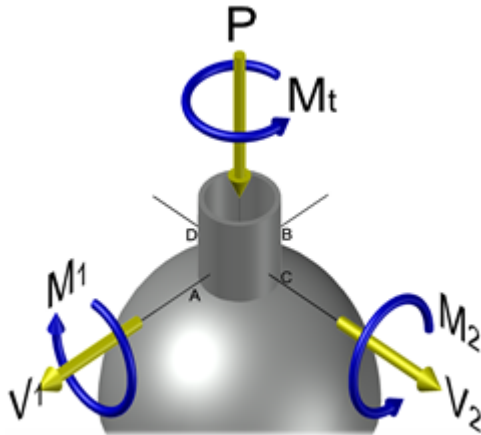
Compute Pressure Stress per WRC-368

No

Local Loads applied at end of Nozzle/Attachment

No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(2629.9 (28.0 - 1.0))}$$

$$= 939.682 \text{ mm.}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load		P	382.4	Kgf
Circumferential Shear	(VC)	V2	254.9	Kgf
Longitudinal Shear	(VL)	V1	254.9	Kgf
Circumferential Moment	(MC)	M1	35.7	Kg-m.
Longitudinal Moment	(ML)	M2	35.7	Kg-m.
Torsional Moment		MT	51.0	Kg-m.

Dimensionless Param: U = 0.11 TAU = 5.73 RHO = 4.00 (5.57)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.10626	(A,B,C,D)
$M(x) / P$	SP 4	0.04132	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.48579	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.20957	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.48579	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.20957	(A,B,C,D)
$N(y) * T / P$	SP 4	0.33054	(A,B,C,D)
$M(y) / P$	SP 4	0.30295	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10321	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	1.66844	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10321	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	1.66844	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Rad. Memb. P	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
Rad. Bend. P	-1.3	1.3	-1.3	1.3	-1.3	1.3	-1.3	1.3
Rad. Memb. MC	0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Rad. Bend. MC	0.0	0.0	0.0	0.0	-2.3	2.3	2.3	-2.3
Rad. Memb. ML	-0.9	-0.9	0.9	0.9	0.0	0.0	0.0	0.0
Rad. Bend. ML	-2.3	2.3	2.3	-2.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-5.0	2.1	1.3	-0.7	-5.0	2.1	1.3	-0.7
Tang. Memb. P	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7
Tang. Bend. P	-9.4	9.4	-9.4	9.4	-9.4	9.4	-9.4	9.4
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang. Bend. MC	0.0	0.0	0.0	0.0	-18.0	18.0	18.0	-18.0
Tang. Memb. ML	-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Tang. Bend. ML	-18.0	18.0	18.0	-18.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-29.3	25.5	7.2	-10.2	-29.3	25.5	7.2	-10.2
Shear VC	1.0	1.0	-1.0	-1.0	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear MT	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Tot. Shear	4.2	4.2	2.3	2.3	2.3	2.3	4.2	4.2
Str. Int.	30.0	26.2	7.9	10.7	29.5	25.7	10.3	12.7

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		29.8	29.8	29.8	29.8	29.8	29.8	29.8	29.8
Rad. Pl (SUS)		-1.4	-1.4	0.3	0.3	-1.4	-1.4	0.3	0.3
Rad. Q (SUS)		-3.5	3.5	1.0	-1.0	-3.5	3.5	1.0	-1.0
Long. Pm (SUS)		29.8	29.8	29.8	29.8	29.8	29.8	29.8	29.8
Long. Pl (SUS)		-1.9	-1.9	-1.5	-1.5	-1.9	-1.9	-1.5	-1.5
Long. Q (SUS)		-27.4	27.4	8.7	-8.7	-27.4	27.4	8.7	-8.7
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.0	1.0	-1.0	-1.0	-1.0	-1.0	1.0	1.0
Shear Q (SUS)		3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Pm (SUS)		29.8	29.8	29.8	29.8	29.8	29.8	29.8	29.8
Pm+Pl (SUS)		29.1	29.1	30.5	30.5	29.1	29.1	30.5	30.5
Pm+Pl+Q (Total)		25.7	56.0	37.7	29.6	25.0	55.5	39.1	30.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	29.76	138.00	Passed
Pm+Pl (SUS)	30.51	207.00	Passed
Pm+Pl+Q (TOTAL)	55.99	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

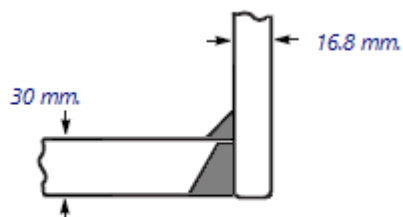
**Input, Nozzle Desc: K1-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.007	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2906.00	mm.
Shell Finished (Minimum) Thickness	t	30.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K1-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----

Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.01 \cdot 1454 / (138 \cdot 1 - 0.6 \cdot 6.01)$
 $= 6.3461$ mm.

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4461 and the shell thk. is 30.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 (\exp([6.01 / (101 \cdot 1)] - 1))$
 $= 0.1511$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	142.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	71.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1511	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 7.3461	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.3461	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 4.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.35, 2.5)]$
 $= 4.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.15, 4.8)$
 $= 4.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K1-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	16.7640	mm.
---	------	---------	-----

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6119 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 236.9869 mm.

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Input, Nozzle Desc: K3-2"

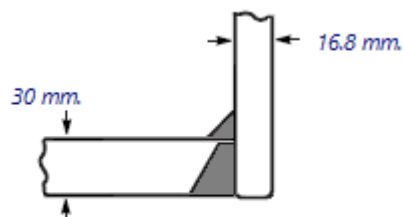
From: 40

Pressure for Reinforcement Calculations	P	6.007	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2906.00	mm.
Shell Finished (Minimum) Thickness	t	30.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		80.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----

Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.01 \cdot 1454 / (138 \cdot 1 - 0.6 \cdot 6.01)$
 $= 6.3461$ mm.

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4461 and the shell thk. is 30.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 (\exp([6.01 / (101 \cdot 1)] - 1))$
 $= 0.1511$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	142.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	71.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1511	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 7.3461	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.3461	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 4.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.35, 2.5)]$
 $= 4.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.15, 4.8)$
 $= 4.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K3-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	16.7640	mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6119 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 236.9869 mm.

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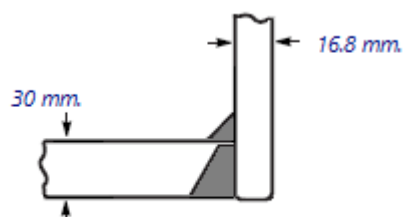

Input, Nozzle Desc: K5-2"
From: 40

Pressure for Reinforcement Calculations	P	6.007	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2906.00	mm.
Shell Finished (Minimum) Thickness	t	30.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		100.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).


Reinforcement CALCULATION, Description: K5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.01 \cdot 1454 / (138 \cdot 1 - 0.6 \cdot 6.01)$
 $= 6.3461$ mm.

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4461 and the shell thk. is 30.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 (\exp([6.01 / (101 \cdot 1)] - 1))$
 $= 0.1511$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	142.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	71.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K5-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1511	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 7.3461	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.3461	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 4.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.35, 2.5)]$
 $= 4.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.15, 4.8)$
 $= 4.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K5-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	16.7640	mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6119 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 236.9869 mm.

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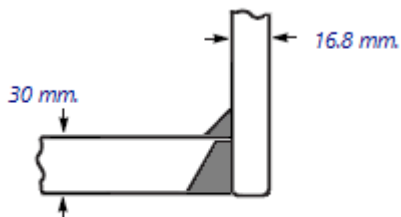
**Input, Nozzle Desc: K7-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.007	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2906.00	mm.
Shell Finished (Minimum) Thickness	t	30.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K7-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.01 \cdot 1454 / (138 \cdot 1 - 0.6 \cdot 6.01)$
 $= 6.3461$ mm.

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4461 and the shell thk. is 30.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 (\exp([6.01 / (101 \cdot 1)] - 1))$
 $= 0.1511$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	142.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	71.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K7-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1511	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 7.3461	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.3461	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 4.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.35, 2.5)]$
 $= 4.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.15, 4.8)$
 $= 4.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K7-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	16.7640	mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6119 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 236.9869 mm.

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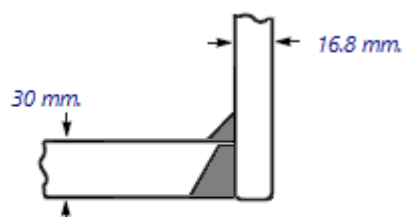
**Input, Nozzle Desc: K2-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.056	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2906.00	mm.
Shell Finished (Minimum) Thickness	t	30.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		80.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K2-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.06 \cdot 1454 / (138 \cdot 1 - 0.6 \cdot 6.06)$
 $= 6.3975$ mm.

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4461 and the shell thk. is 30.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 (\exp([6.06 / (101 \cdot 1)] - 1))$
 $= 0.1523$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	142.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	71.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1523	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 7.3975	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.3975	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 4.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.4, 2.5)]$
 $= 4.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.15, 4.8)$
 $= 4.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	16.7640	mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6119 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 236.9869 mm.

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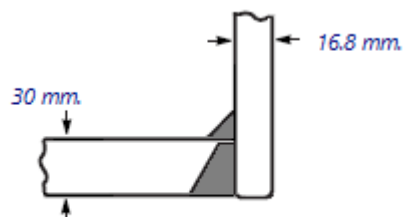

Input, Nozzle Desc: K4-2"
From: 40

Pressure for Reinforcement Calculations	P	6.056	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2906.00	mm.
Shell Finished (Minimum) Thickness	t	30.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).


Reinforcement CALCULATION, Description: K4-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.06 \cdot 1454 / (138 \cdot 1 - 0.6 \cdot 6.06)$
 $= 6.3975$ mm.

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4461 and the shell thk. is 30.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 (\exp([6.06 / (101 \cdot 1)] - 1))$
 $= 0.1523$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	142.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	71.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1523	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 7.3975	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.3975	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 4.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.4, 2.5)]$
 $= 4.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.15, 4.8)$
 $= 4.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K4-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	16.7640	mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6119 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 236.9869 mm.

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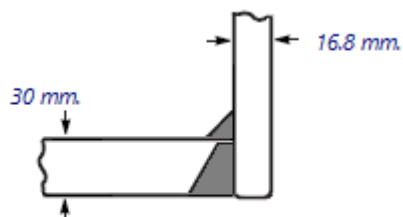
**Input, Nozzle Desc: K6-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.056	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2906.00	mm.
Shell Finished (Minimum) Thickness	t	30.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		100.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K6-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----



Actual Thickness Used in Calculation

0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.06 \cdot 1454 / (138 \cdot 1 \cdot 0.6 \cdot 6.06)$
 $= 6.3975 \text{ mm.}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4461 and the shell thk. is 30.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \left(\exp \left(\frac{P}{S \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
 $= 25.4 \left(\exp \left(\frac{6.06}{101 \cdot 1} \right) - 1 \right)$
 $= 0.1523 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	142.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	71.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K6-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1523	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 7.3975	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.3975	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 4.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.4, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.15, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K6-2"

Intermediate Calc. for nozzle/shell Welds Tmin 16.7640 mm.

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6119 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 236.9869 mm.

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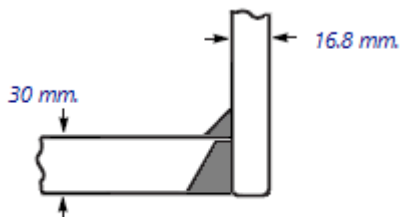
**Input, Nozzle Desc: K8-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.056	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2906.00	mm.
Shell Finished (Minimum) Thickness	t	30.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		110.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K8-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----



Actual Thickness Used in Calculation

0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.06 \cdot 1454 / (138 \cdot 1 \cdot 0.6 \cdot 6.06)$
 $= 6.3975 \text{ mm.}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4461 and the shell thk. is 30.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \left(\exp \left(\frac{P}{S \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
 $= 25.4 \left(\exp \left(\frac{6.06}{101 \cdot 1} \right) - 1 \right)$
 $= 0.1523 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	142.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	71.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K8-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1523	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 7.3975	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.3975	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 4.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.4, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.15, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K8-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	16.7640	mm.
---	------	---------	-----

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6119 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 236.9869 mm.

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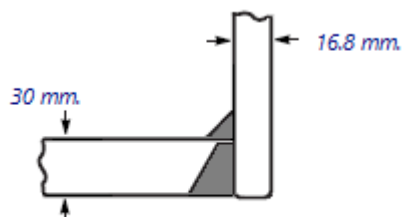
**Input, Nozzle Desc: K10-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.056	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2906.00	mm.
Shell Finished (Minimum) Thickness	t	30.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K10-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.



Actual Thickness Used in Calculation

0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.06 \cdot 1454 / (138 \cdot 1 - 0.6 \cdot 6.06)$
 $= 6.3975 \text{ mm.}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4461 and the shell thk. is 30.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
 $= 25.4 \left(\exp \left(\frac{6.06}{101 \cdot 1} \right) - 1 \right)$
 $= 0.1523 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	142.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	71.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K10-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1523	mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 7.3975	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.3975	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm.
Wall Thickness per table UG-45	tb3	= 4.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.4, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.15, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K10-2"

Intermediate Calc. for nozzle/shell Welds Tmin 16.7640 mm.

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6119 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 236.9869 mm.

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Input, Nozzle Desc: N1-24"

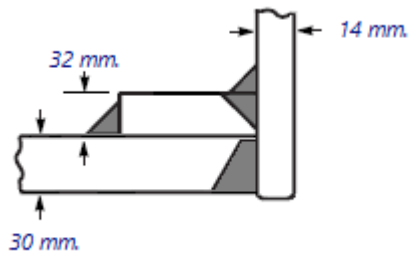
From: 50

Pressure for Reinforcement Calculations	P	6.007	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2906.00	mm.
Shell Finished (Minimum) Thickness	t	30.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1750.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1109.6000	mm.
Thickness of Pad	te	32.0000	mm.
Weld leg size between Pad and Shell	Wp	24.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm.
Reinforcing Pad Width		250.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N1-24"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.551 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.01 \cdot 1454 / (138 \cdot 1 - 0.6 \cdot 6.01)$
 $= 6.3461 \text{ mm.}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4467 and the shell thk. is 30.0000 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.01 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 6.01)$
 $= 1.3245 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1175.2001 mm.
Parallel to Vessel Wall, opening length	d	587.6000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		59.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

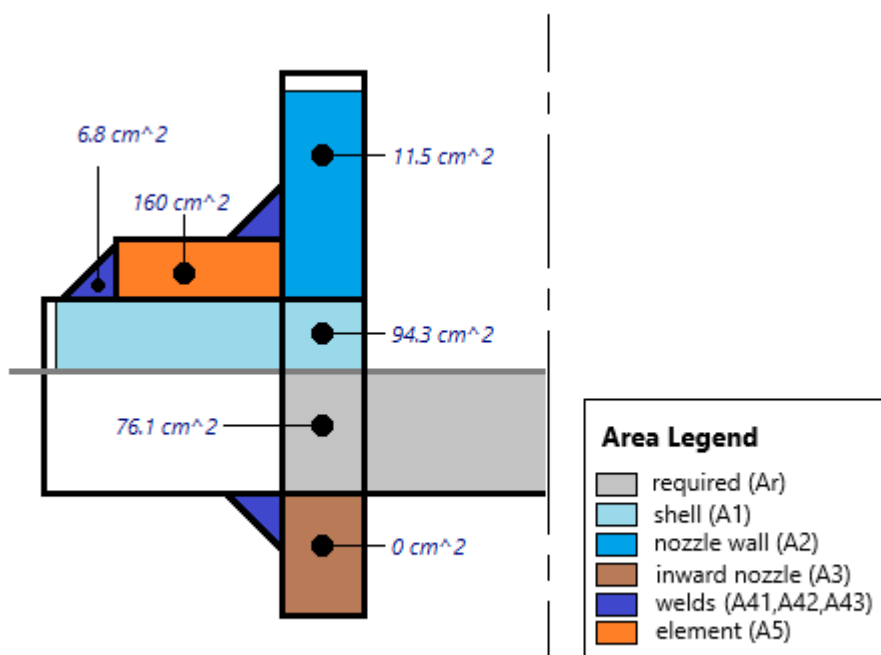


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	76.123	NA	NA
Area in Shell	A1	94.281	NA	NA
Area in Nozzle Wall	A2	11.514	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	6.760	NA	NA
Area in Element	A5	160.000	NA	NA
TOTAL AREA AVAILABLE	Atot	272.555	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (588 \cdot 13 \cdot 1 + 2 \cdot 11 \cdot 13 \cdot 1 \cdot (1 - 1))$
 $= 76.123 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 588 (1 \cdot 29 - 1 \cdot 13) - 2 \cdot 11$
 $(1 \cdot 29 - 1 \cdot 13) \cdot (1 - 1)$
 $= 94.281 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 59.5) \cdot (11 - 1.32) \cdot 1$
 $= 11.514 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 24^2 \cdot 1$$

$$= 6.760 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot f_{r4}$$

$$= (1110 - 610) \cdot 32 \cdot 1$$

$$= 160.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.3245 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 7.3461 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 7.3461 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(7.35, 4.5)]$$

$$= 7.3461 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.32, 7.35)$$

$$= 7.3461 \text{ mm.}$$

Available Nozzle Neck Thickness = 14.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	33.5,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	311.5 N./mm ²	Passed
Occasional	:	7.9,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	16.2,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, tg = 14, tr = 1.32, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.12, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, tg = 14, tr = 1.32, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.12, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 14, tr = 1.32, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.12, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 30, tr = 6.35, c = 1 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.22, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve D -30 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, $t_g = 14$, $t_r = 1.32$, $c = 3$ mm., $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.01/51.00 = 0.118$

Weld Size Calculations, Description: N1-24"

Intermediate Calc. for nozzle/shell Welds T_{min} 11.0000 mm.
 Intermediate Calc. for pad/shell Welds T_{minPad} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$16.9680 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (76.1 - 94.3 + 2 \cdot 11 \cdot 1 \cdot (1 \cdot 29 - 13)) \cdot 138)$
 $= \max(0, -20584.45) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - C_{an}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (11.5 + 160 + 6.76 - 0 \cdot 1) \cdot 138$
 $= 250863.81 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (11.5 + 0 + 1 + (6.38)) \cdot 138$
 $= 26587.03 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (11.5 + 0 + 6.76 + 160 + (6.38)) \cdot 138$
 $= 259841.64 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$
 $= 66025. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 1110 \cdot 24 \cdot 0.49 \cdot 138$



= 288432. Kgf

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 299) \cdot (14 - 3) \cdot 0.7 \cdot 138$$

$$= 101882. Kgf$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 610 \cdot 32 \cdot 0.74 \cdot 138$$

$$= 319078. Kgf$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 610 \cdot (30 - 1) \cdot 0.74 \cdot 138$$

$$= 289164. Kgf$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (288432 + 101882) = 390314 Kgf$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (66025 + 319078 + 289164 + 0) = 674268 Kgf$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (288432 + 289164 + 0) = 577597 Kgf$$

Summary of Failure Path Calculations:

Path 1-1 = 390314 Kgf, must exceed W = 0 Kgf	or W1 = 250863 Kgf
Path 2-2 = 674268 Kgf, must exceed W = 0 Kgf	or W2 = 26587 Kgf
Path 3-3 = 577596 Kgf, must exceed W = 0 Kgf	or W3 = 259841 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 32.3290 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 362.3290 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.351 %

Input Echo, WRC297 Item 1, Description: N1-24" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2906.000	mm.
Shell Thickness	30.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	609.600	mm.
Nozzle Thickness	14.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	



Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1109.600	mm.
Thickness of Reinforcing Pad	32.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	6.01	bars
Radial Load	P	4589.03	Kgf
Circumferential Shear	Vc	3059.35	Kgf
Longitudinal Shear	Vl	3059.35	Kgf
Circumferential Moment	Mc	5137.73	Kg-m.
Longitudinal Moment	Ml	5137.73	Kg-m.
Torsional Moment	Mt	7339.62	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 609.6 + 2 \cdot 1.65 \cdot \sqrt{(1484.5(30.0 - 1.0))}$$

$$= 1294.304 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2969.000	mm.
Nozzle Outside Diameter	(d) =	609.600	mm.
Vessel Thickness used	(T) =	61.000	mm.
Nozzle Thickness used	(t) =	11.000	mm.
T / t	=	5.545	
d / t	=	55.418	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.432$$

Nr/P	=	0.035
Mr/P	=	0.031
M0/P	=	0.086
N0/P	=	0.298
MrD/Mc	=	0.072
NrDL/Mc	=	0.071
M0d/Mc	=	0.240
N0DL/Mc	=	0.366
MrD/Ml	=	0.055
NrDL/Ml	=	0.055
M0D/Ml	=	0.150
N0DL/Ml	=	0.600

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-13	-13	13	13
Outplane Bending (Ml)	-20	20	20	-20
Normal Pressure Stress	14	14	14	14



Outplane Stress Summary	-29	23	37	10
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-2	2	-2	2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-7	7	7	-7
Inplane Pressure Stress	7	7	7	7

Inplane Stress Summary	-4	14	12	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	29	24	37	10
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-31	31	31	-31
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	7	7	7	7

Outplane Stress Summary	-42	33	37	-14
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-2	2	-2	2
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-9	9	9	-9
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	14	14	14	14

Inplane Stress Summary	0	24	22	8
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to	Cu Left	Cl Left	Du Right	Dl Right
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circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	43	33	37	23

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	19	18	19
Hoop Stress Summary	1	2	28	29

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-30	30	-30	30
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-16	-16	16	16
Axial Bending (ML)	-112	112	112	-112
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	-152	133	105	-57

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11
Shear Stress Summary	14	14	8	8

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	156	135	106	88

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	19	18	19

Hoop Stress Summary	7	7	23	24
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-30	30	-30	30
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-149	149	149	-149
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9

Axial Stress Summary	-189	170	142	-95
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	11	11	11	11

Shear Stress Summary	8	8	14	14
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	197	171	144	122
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 609.6 + 2 \cdot 1.65 \cdot \sqrt{(1468.5(30.0 - 1.0))}$$

$$= 1290.604 \text{ mm.}$$

WRC 297 Curve Access Parameters:

$$\text{Vessel Mean Diameter (D)} = 2937.000 \text{ mm.}$$

$$\text{Nozzle Outside Diameter (d)} = 1109.600 \text{ mm.}$$

$$\text{Vessel Thickness used (T)} = 29.000 \text{ mm.}$$

$$\text{Nozzle Thickness used (t)} = 261.000 \text{ mm.}$$

$$T / t = 0.111$$

$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.802$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

$$Nr/P = 0.053$$

$$Mr/P = 0.087$$

$$M0/P = 0.025$$

$$N0/P = 0.038$$

$$MrD/Mc = 0.256$$

$$NrDL/Mc = 0.103$$

$$M0d/Mc = 0.076$$

$$N0DL/Mc = 0.064$$

$$MrD/Ml = 0.065$$

$$NrDL/Ml = 0.062$$



MØD/M1 = 0.019

NØDL/M1 = 0.043

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-8	8	-8	8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	29	30	29	30
Outplane Stress Summary	11	40	28	32

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-28	28	-28	28
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-20	20	20	-20
Inplane Pressure Stress	14	14	14	14
Inplane Stress Summary	-40	57	8	22

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	51	57	28	32

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-8	8	-8	8
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-24	24	24	-24
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	14	14	14	14
Outplane Stress Summary	-23	42	32	0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to	Cu Left	Cl Left	Du Right	Dl Right
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circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-28	28	-28	28
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-82	82	82	-82
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	29	30	29	30
Inplane Stress Summary	-89	132	87	-21

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	89	132	87	21

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	14	14	14	14	14	14	14	14
Circ. Pl (SUS)	-16	-16	10	10	-1	-1	1	1
Circ. Q (SUS)	-26	26	14	-14	-11	11	7	-7
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-1	-1	1	1	-11	-11	5	5
Long. Q (SUS)	-9	9	5	-5	-37	37	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	14.0	14.0	14.0	14.0	14.0	14.0	14.0	14.0
Pm+Pl (SUS)	8.0	8.0	24.0	24.0	17.0	17.0	15.0	15.0
Pm+Pl+Q (Total)	28.2	24.4	38.2	10.5	43.2	33.4	37.3	21.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	138.00	Passed
Pm+Pl (SUS)	24.00	207.00	Passed
Pm+Pl+Q (TOTAL)	43.19	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	29	30	29	30	29	30	29	30
Circ. Pl (SUS)	-4	-4	0	0	-7	-7	3	3
Circ. Q (SUS)	-14	14	-2	2	-110	110	54	-54
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-5	-5	1	1	-5	-5	1	1
Long. Q (SUS)	-48	48	-8	8	-32	32	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	29.0	30.0	29.0	30.0	29.0	30.0	29.0	30.0
Pm+Pl (SUS)	25.0	26.0	29.0	30.0	22.0	23.0	32.0	33.0
Pm+Pl+Q (Total)	50.0	57.1	27.0	32.1	88.0	133.0	86.0	21.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30.00	138.00	Passed
Pm+Pl (SUS)	33.00	207.00	Passed
Pm+Pl+Q (TOTAL)	133.01	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18	19	18	19	18	19	18	19
Circ. Pl (SUS)	-16	-16	10	10	-11	-11	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-18	-18	14	14	-18	-18	14	14
Long. Q (SUS)	-142	142	82	-82	-179	179	119	-119
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	18.0	19.0	18.0	19.0	18.0	19.0	18.0	19.0
Pm+Pl (SUS)	11.7	12.6	28.7	29.6	16.5	17.5	25.0	25.6
Pm+Pl+Q (Total)	155.2	134.3	106.0	89.8	195.8	170.5	143.4	122.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	138.00	Passed

Process Air 3rd KOD 198-D

PV Elite 28 U1 Licensee: SL User



FileName : Process Air Compressor 3rd KO Drum-198-D-7%

Page 122 of 145

Nozzle Calcs.: N1-24"

Nozl: 39 2:30pm Sep 16,2026



HEXAGON

Pm+Pl (SUS)	29.61	207.00	Passed
Pm+Pl+Q (TOTAL)	195.83	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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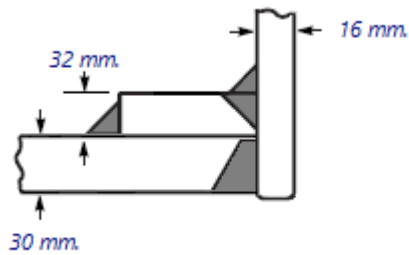
Input, Nozzle Desc: N2-24"
From: 50

Pressure for Reinforcement Calculations	P	6.007	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2906.00	mm.
Shell Finished (Minimum) Thickness	t	30.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2350.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1109.6000	mm.
Thickness of Pad	te	32.0000	mm.
Weld leg size between Pad and Shell	Wp	24.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm.
Reinforcing Pad Width		250.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N2-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.630 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.01 \cdot 1454 / (138 \cdot 1 - 0.6 \cdot 6.01)$
 $= 6.3461 \text{ mm.}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4467 and the shell thk. is 30.0000 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.01 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 6.01)$
 $= 1.3245 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1167.2001 mm.
Parallel to Vessel Wall, opening length	d	583.6000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		64.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

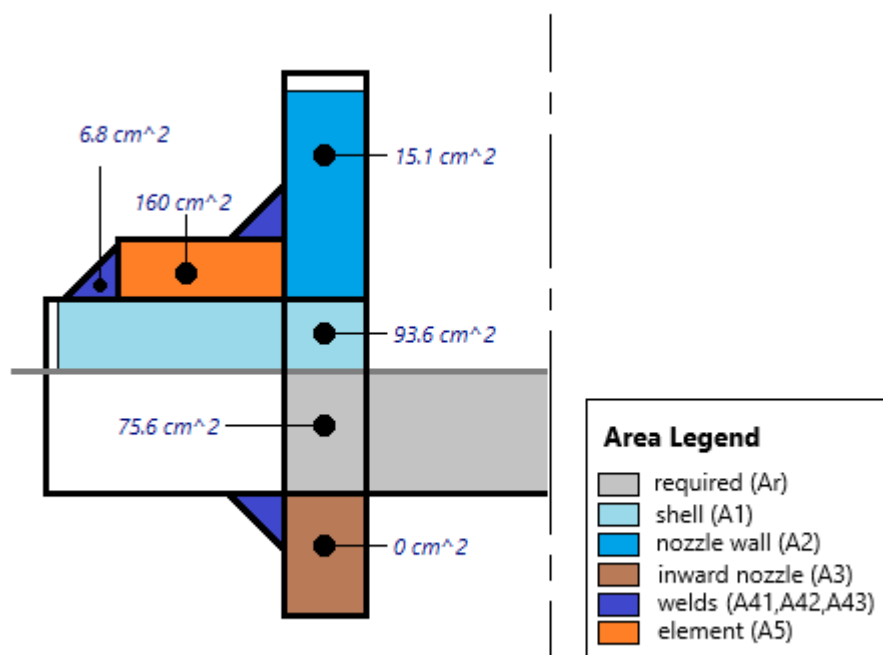


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	75.605	NA	NA
Area in Shell	A1	93.639	NA	NA
Area in Nozzle Wall	A2	15.061	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	6.760	NA	NA
Area in Element	A5	160.000	NA	NA
TOTAL AREA AVAILABLE	Atot	275.461	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (584 \cdot 13 \cdot 1 + 2 \cdot 13 \cdot 13 \cdot 1 \cdot (1 - 1))$
 $= 75.605 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 584 (1 \cdot 29 - 1 \cdot 13) - 2 \cdot 13$
 $(1 \cdot 29 - 1 \cdot 13) \cdot (1 - 1)$
 $= 93.639 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 64.5) \cdot (13 - 1.32) \cdot 1$
 $= 15.061 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$



$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 24^2 \cdot 1$$

$$= 6.760 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot f_{r4}$$

$$= (1110 - 610) \cdot 32 \cdot 1$$

$$= 160.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.3245 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 7.3461 \text{ mm.}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 7.3461 \text{ mm.}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(7.35, 4.5)]$$

$$= 7.3461 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.32, 7.35)$$

$$= 7.3461 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	24.8,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	320.2 N./mm ²	Passed
Occasional	:	6.6,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	3.6,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

$$\text{Govrn. thk, } t_g = 16, tr = 1.32, c = 3 \text{ mm., } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.1, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

$$\text{Govrn. thk, } t_g = 16, tr = 1.32, c = 3 \text{ mm., } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.1, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve D	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

$$\text{Govrn. thk, } t_g = 16, tr = 1.32, c = 3 \text{ mm., } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.1, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve D	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

$$\text{Govrn. thk, } t_g = 30, tr = 6.35, c = 1 \text{ mm., } E^* = 1$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.22, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$



Min Metal Temp. w/o impact per UCS-66, Curve D -30 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, $t_g = 16$, $t_r = 1.32$, $c = 3$ mm., $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -46 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.01/51.00 = 0.118$

Weld Size Calculations, Description: N2-24"

Intermediate Calc. for nozzle/shell Welds T_{min} 13.0000 mm.
 Intermediate Calc. for pad/shell Welds T_{minPad} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$16.9680 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (75.6 - 93.6 + 2 \cdot 13 \cdot 1 \cdot (1 \cdot 29 - 13)) \cdot 138)$
 $= \max(0, -19507.37) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - C_{an}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (15.1 + 160 + 6.76 - 0 \cdot 1) \cdot 138$
 $= 255855.80 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (15.1 + 0 + 1 + (7.54)) \cdot 138$
 $= 33211.40 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (15.1 + 0 + 6.76 + 160 + (7.54)) \cdot 138$
 $= 266465.94 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$
 $= 66025. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 1110 \cdot 24 \cdot 0.49 \cdot 138$



= 288432. Kgf

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 298) \cdot (16 - 3) \cdot 0.7 \cdot 138$$

$$= 120004. Kgf$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 610 \cdot 32 \cdot 0.74 \cdot 138$$

$$= 319078. Kgf$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 610 \cdot (30 - 1) \cdot 0.74 \cdot 138$$

$$= 289164. Kgf$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (288432 + 120004) = 408436 Kgf$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (66025 + 319078 + 289164 + 0) = 674268 Kgf$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (288432 + 289164 + 0) = 577597 Kgf$$

Summary of Failure Path Calculations:

Path 1-1 = 408436 Kgf, must exceed W = 0 Kgf	or W1 = 255855 Kgf
Path 2-2 = 674268 Kgf, must exceed W = 0 Kgf	or W2 = 33211 Kgf
Path 3-3 = 577596 Kgf, must exceed W = 0 Kgf	or W3 = 266465 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 32.3290 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 362.3290 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.695 %

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 294.29 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 583.60 mm.

Input Echo, WRC297 Item 1, Description: N2-24" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2906.000	mm.
Shell Thickness	30.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm2
Vessel Hot S.I. Allowable	Smh 138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.



Nozzle Diameter	609.600	mm.
Nozzle Thickness	16.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1109.600	mm.
Thickness of Reinforcing Pad	32.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	6.01	bars
Radial Load	P	-4589.03	Kgf
Circumferential Shear	Vc	-3059.35	Kgf
Longitudinal Shear	Vl	-3059.35	Kgf
Circumferential Moment	Mc	-5137.73	Kg-m.
Longitudinal Moment	Ml	-5137.73	Kg-m.
Torsional Moment	Mt	-7339.62	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 609.6 + 2 \cdot 1.65 \cdot \sqrt{(1484.5(30.0 - 1.0))}$$

$$= 1294.304 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2969.000	mm.
Nozzle Outside Diameter	(d) =	609.600	mm.
Vessel Thickness used	(T) =	61.000	mm.
Nozzle Thickness used	(t) =	13.000	mm.
T / t	=	4.692	
d / t	=	46.892	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.432$$

Nr/P	=	0.043
Mr/P	=	0.043
M0/P	=	0.082
N0/P	=	0.275
MrD/Mc	=	0.091
NrDL/Mc	=	0.082
M0d/Mc	=	0.225
N0DL/Mc	=	0.327
MrD/Ml	=	0.069
NrDL/Ml	=	0.064
M0D/Ml	=	0.142
N0DL/Ml	=	0.541

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0



Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	12	12	-12	-12
Outplane Bending (ML)	18	-18	-18	18
Normal Pressure Stress	14	14	14	14

Outplane Stress Summary	54	5	-7	18
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	1	1	-1	-1
Inplane Bending (ML)	9	-9	-9	9
Inplane Pressure Stress	7	7	7	7

Inplane Stress Summary	21	-3	0	12
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	-2	-2	-2	-2

Shear Stress Summary	-2	-2	-1	-1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	54	9	8	19
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	7	7	-7	-7
Outplane Bending (Mc)	29	-29	-29	29
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	7	7	7	7

Outplane Stress Summary	53	-18	-20	27
-------------------------	----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	1	1	-1	-1
Inplane Bending (Mc)	12	-12	-12	12
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	14	14	14	14

Inplane Stress Summary	31	1	3	22
------------------------	----	---	---	----

**Vessel Stresses (N./mm²):**

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		-2	-2	-2	-2
Shear Stress Summary		-1	-1	-2	-2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		53	20	25	28

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses in the hoop direction)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)		3	3	3	3
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		12	12	-12	-12
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		15	16	15	16
Hoop Stress Summary		31	31	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)		1	1	1	1
Axial Bending (P)		33	-33	33	-33
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		14	14	-14	-14
Axial Bending (ML)		108	-108	-108	108
Axial Pressure Stress		7	7	7	7
Axial Stress Summary		166	-118	-78	69

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress)		Outside	Inside	Outside	Inside
Shear due to (Vc)		-2	-2	2	2
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		-9	-9	-9	-9
Shear Stress Summary		-11	-11	-7	-7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Outside	Inside	Outside	Inside
Two * Max Shear Stress		167	152	86	70

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses in the hoop direction)		Left Outside	Left Inside	Right Outside	Right Inside



Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	7	7	-7	-7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	16	15	16
Hoop Stress Summary	26	27	11	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	33	-33	33	-33
Axial Membrane (Mc)	14	14	-14	-14
Axial Bending (Mc)	147	-147	-147	147
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	205	-157	-118	109

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	2	2	-2	-2
Shear due to Torsion	-9	-9	-9	-9
Shear Stress Summary	-7	-7	-11	-11

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	205	185	132	110

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 609.6 + 2 \cdot 1.65 \cdot \sqrt{1468.5(30.0 - 1.0)}$$

$$= 1290.604 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2937.000 mm.
Nozzle Outside Diameter	(d) =	1109.600 mm.
Vessel Thickness used	(T) =	29.000 mm.
Nozzle Thickness used	(t) =	263.000 mm.
T / t	=	0.110
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.802$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.053
Mr/P	=	0.087
M0/P	=	0.025
N0/P	=	0.038
MrD/Mc	=	0.256
NrDL/Mc	=	0.103



$M\theta d/Mc = 0.076$
 $N\theta DL/Mc = 0.063$
 $MrD/Ml = 0.065$
 $NrDL/Ml = 0.062$
 $M\theta D/Ml = 0.019$
 $N\theta DL/Ml = 0.043$

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	2	2	-2	-2
Outplane Bending (ML)	6	-6	-6	6
Normal Pressure Stress	29	30	29	30
Outplane Stress Summary	48	20	31	28

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	28	-28	28	-28
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	3	3	-3	-3
Inplane Bending (ML)	20	-20	-20	20
Inplane Pressure Stress	14	14	14	14
Inplane Stress Summary	70	-27	21	7

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	-1	-1	-1	-1
Shear Stress Summary	-1	-1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	70	48	31	28

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	3	3	-3	-3
Outplane Bending (Mc)	24	-24	-24	24
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	14	14	14	14
Outplane Stress Summary	53	-12	-3	30

**Vessel Stresses (N./mm²):**

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	28	-28	28	-28
Inplane Membrane (Mc)	5	5	-5	-5
Inplane Bending (Mc)	82	-82	-82	82
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	29	30	29	30
Inplane Stress Summary	149	-72	-27	82

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	-1	-1	-1	-1
Shear Stress Summary	0	0	-1	-1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	149	72	27	82

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	14	14	14	14	14	14	14	14
Circ. Pl (SUS)	15	15	-9	-9	1	1	-1	-1
Circ. Q (SUS)	23	-23	-13	13	15	-15	-9	9
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	1	1	-1	-1	10	10	-4	-4
Long. Q (SUS)	12	-12	-6	6	34	-34	-24	24
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	-2	-2	-2	-2	-2	-2	-2	-2
Pm (SUS)	14.0	14.0	14.0	14.0	14.0	14.0	14.0	14.0
Pm+Pl (SUS)	29.0	29.0	6.0	6.0	17.0	17.0	13.0	13.0
Pm+Pl+Q (Total)	52.1	10.8	8.9	18.6	51.2	17.5	25.3	27.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	138.00	Passed
Pm+Pl (SUS)	29.00	207.00	Passed
Pm+Pl+Q (TOTAL)	52.12	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	29	30	29	30	29	30	29	30
Circ. Pl (SUS)	4	4	0	0	7	7	-3	-3
Circ. Q (SUS)	14	-14	2	-2	110	-110	-54	54
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	5	5	-1	-1	5	5	-1	-1
Long. Q (SUS)	48	-48	8	-8	32	-32	-16	16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	-1	-1	-1	-1	-1	-1	-1	-1
Pm (SUS)	29.0	30.0	29.0	30.0	29.0	30.0	29.0	30.0
Pm+Pl (SUS)	33.0	34.0	29.0	30.0	36.0	37.0	26.0	27.0
Pm+Pl+Q (Total)	67.0	49.0	31.1	28.0	146.0	73.0	28.0	81.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30.00	138.00	Passed
Pm+Pl (SUS)	37.00	207.00	Passed
Pm+Pl+Q (TOTAL)	146.01	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	16	15	16	15	16	15	16
Circ. Pl (SUS)	15	15	-9	-9	10	10	-4	-4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	15	15	-13	-13	15	15	-13	-13
Long. Q (SUS)	141	-141	-75	75	180	-180	-114	114
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	-2	-2	2	2	2	2	-2	-2
Shear Q (SUS)	-9	-9	-9	-9	-9	-9	-9	-9
Pm (SUS)	15.0	16.0	15.0	16.0	15.0	16.0	15.0	16.0
Pm+Pl (SUS)	30.5	31.4	12.6	13.6	26.0	26.8	17.5	18.4
Pm+Pl+Q (Total)	163.9	151.6	88.1	69.8	202.3	184.5	132.8	109.2

Nozzle Stress Summation Comparison (N./mm²):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16.00	138.00	Passed
Pm+Pl (SUS)	31.42	207.00	Passed
Pm+Pl+Q (TOTAL)	202.28	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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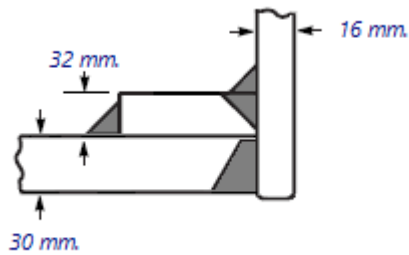
**Input, Nozzle Desc: M1-24"****From: 50**

Pressure for Reinforcement Calculations	P	6.007	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2906.00	mm.
Shell Finished (Minimum) Thickness	t	30.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1500.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1109.6000	mm.
Thickness of Pad	te	32.0000	mm.
Weld leg size between Pad and Shell	Wp	24.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm.
Reinforcing Pad Width		250.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: M1-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.630 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.01 \cdot 1454 / (138 \cdot 1 - 0.6 \cdot 6.01)$
 $= 6.3461 \text{ mm.}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4467 and the shell thk. is 30.0000 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.01 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 6.01)$
 $= 1.3245 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1167.2001 mm.
Parallel to Vessel Wall, opening length	d	583.6000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		64.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

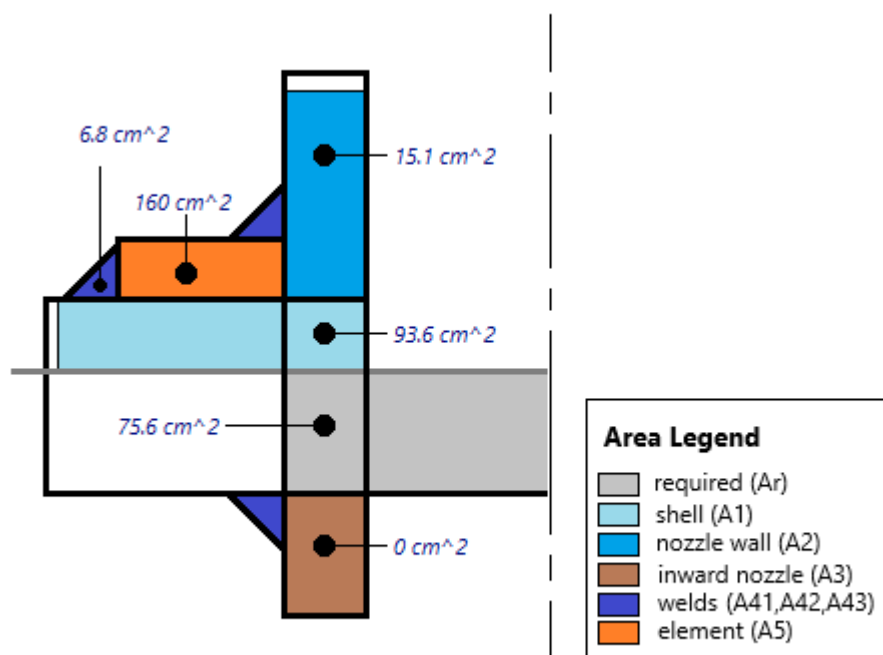


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	75.605	NA	NA
Area in Shell	A1	93.639	NA	NA
Area in Nozzle Wall	A2	15.061	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	6.760	NA	NA
Area in Element	A5	160.000	NA	NA
TOTAL AREA AVAILABLE	Atot	275.461	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (584 \cdot 13 \cdot 1 + 2 \cdot 13 \cdot 13 \cdot 1 \cdot (1 - 1))$$

$$= 75.605 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 584 (1 \cdot 29 - 1 \cdot 13) - 2 \cdot 13$$

$$(1 \cdot 29 - 1 \cdot 13) \cdot (1 - 1)$$

$$= 93.639 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 64.5) \cdot (13 - 1.32) \cdot 1$$

$$= 15.061 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$



$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 24^2 \cdot 1$$

$$= 6.760 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot f_{r4}$$

$$= (1110 - 610) \cdot 32 \cdot 1$$

$$= 160.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:Wall Thickness for Internal/External pressures $t_a = 4.3245 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 4.3245 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)**Govrn. thk, $t_g = 16$, $t_r = 1.32$, $c = 3 \text{ mm.}$, $E^* = 1$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: DGovrn. thk, $t_g = 16$, $t_r = 1.32$, $c = 3 \text{ mm.}$, $E^* = 1$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: DGovrn. thk, $t_g = 16$, $t_r = 1.32$, $c = 3 \text{ mm.}$, $E^* = 1$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: DGovrn. thk, $t_g = 30$, $t_r = 6.35$, $c = 1 \text{ mm.}$, $E^* = 1$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.22$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-30 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: DGovrn. thk, $t_g = 16$, $t_r = 1.32$, $c = 3 \text{ mm.}$, $E^* = 1$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :



Design Pressure/Ambient Rating = $6.01/51.00 = 0.118$

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 13.0000 \text{ mm.}$
 Intermediate Calc. for pad/shell Welds $T_{minPad} \quad 19.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$16.9680 = 0.7 * W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (75.6 - 93.6 + 2 \cdot 13 \cdot 1 \cdot (1 \cdot 29 - 13)) \cdot 138)$
 $= \max(0, -19507.37) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (15.1 + 160 + 6.76 - 0 \cdot 1) \cdot 138$
 $= 255855.80 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (15.1 + 0 + 1 + (7.54)) \cdot 138$
 $= 33211.40 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (15.1 + 0 + 6.76 + 160 + (7.54)) \cdot 138$
 $= 266465.94 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$
 $= 66025. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 1110 \cdot 24 \cdot 0.49 \cdot 138$
 $= 288432. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 298) \cdot (16 - 3) \cdot 0.7 \cdot 138$
 $= 120004. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 610 \cdot 32 \cdot 0.74 \cdot 138$
 $= 319078. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 610 \cdot (30 - 1) \cdot 0.74 \cdot 138$
 $= 289164. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:



$$= (\text{SPEW} + \text{SNW}) = (288432 + 120004) = 408436 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (66025 + 319078 + 289164 + 0) = 674268 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (288432 + 289164 + 0) = 577597 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 408436 Kgf, must exceed W = 0 Kgf	or W1 = 255855 Kgf
Path 2-2 = 674268 Kgf, must exceed W = 0 Kgf	or W2 = 33211 Kgf
Path 3-3 = 577596 Kgf, must exceed W = 0 Kgf	or W3 = 266465 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 32.3290 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 362.3290 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 * t_{nom} / R_f * (1 - R_f / R_o)$) 2.695 %

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 423.83 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 583.60 mm.

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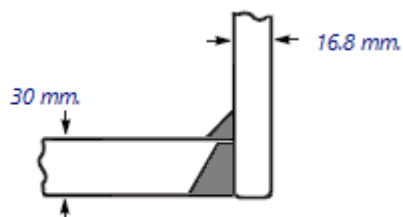

Input, Nozzle Desc: K9-2"
From: 50

Pressure for Reinforcement Calculations	P	6.007	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2906.00	mm.
Shell Finished (Minimum) Thickness	t	30.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2500.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		110.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).


Reinforcement CALCULATION, Description: K9-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----



Actual Thickness Used in Calculation

0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.01 \cdot 1454 / (138 \cdot 1 - 0.6 \cdot 6.01)$
 $= 6.3461 \text{ mm.}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4467 and the shell thk. is 30.0000 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \left(\exp \left(\frac{P}{S \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
 $= 26.4 \left(\exp \left(\frac{6.01}{138 \cdot 1} \right) - 1 \right)$
 $= 0.1152 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	142.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	71.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.4100 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K9-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.1152 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 7.3461 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.3461 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 5.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(7.35, 2.5)]$
 $= 5.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.12, 5.8)$
 $= 5.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D**

Govrn. thk, tg = 16.8, tr = 0.12, c = 1 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.0073$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-44 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Weld Size Calculations, Description: K9-2"

Intermediate Calc. for nozzle/shell Welds T_{min} 15.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6119 mm.

The Cut Length for this Nozzle is, Drop + H_o + H + T : 236.9869 mm.

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**PV Elite[®] Pressure Vessel Design Notes**

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: